DRAWDOWN.

DRAWDOWN.TRADING

DIAGNOSE · MEASURE · PROVE · SCALE

THE EDGE MANUAL

Why you're not consistent yet — and the analytical framework professionals use to find out whether they have an edge, prove it, protect it, and scale it.

21 CHAPTERS / THE SIX PLATEAUS / ANALYTICS SYSTEM / TOOLKIT

> Most traders don't have an edge problem.
> They have a measurement problem — and they've never known the difference.
>
> THE PREMISE OF THIS BOOK

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Risk warning. Trading leveraged products carries a high risk of losing money rapidly, and the majority of retail investor accounts lose money. Past performance and historical testing are not reliable indicators of future results.

Not advice. This book is educational material only. It is not investment advice, a personal recommendation, or a promise of any outcome. It contains no trade recommendations and no strategies to copy. All figures, systems and worked examples are illustrative and constructed to demonstrate method. Statistical models described here rest on assumptions that markets routinely violate — they are tools for thinking, not guarantees. Always verify current rules, costs and regulations with the relevant authority, and consider taking regulated advice before acting.

CONTENTS

What's inside

How to use this book

This book assumes you can already trade. You understand orders, structure and risk; you've had winning weeks and losing months; and you're stuck somewhere between "I know what I'm doing" and "I can rely on this." That gap is the subject of the entire book, and it is an analytical problem far more often than it is a knowledge problem.

You need data to use this properly

This is not a book you can read passively. Almost every chapter asks you to look at your own trades. If you have fewer than 50 recorded trades with entry, stop, size, result in R and a plan-adherence flag, start recording now and read Parts I and II while your sample builds. If you have no records at all, that is itself the diagnosis — go to Chapter 3, then come back.

Three reading paths

YOUR SITUATION	START HERE
"I keep switching strategies"	Chapters 2, 4 and 6. You almost certainly abandoned at least one workable edge inside its normal variance. The maths in Chapter 6 is the antidote and it will probably annoy you.
"My results are wildly inconsistent"	Chapters 3, 15 and 16. Inconsistent aggregate results usually conceal a stable pattern — one setup or session funding the losses of the others. Segmentation finds it.
"I was profitable, now I'm not"	Chapters 4, 13 and 16. Three candidates: normal variance, genuine edge decay, or execution drift. They look identical from the equity curve and completely different in the data.

On the mathematics

There is real statistics in Part II — standard error, confidence intervals, Monte Carlo simulation, the Kelly criterion. None of it requires more than arithmetic and a spreadsheet, and all of it is explained from first principles. It's here because these concepts are the difference between knowing whether you have an edge and merely believing it, and that distinction is the entire book.

> DRAWDOWN INSIGHT
>
> Every simulation figure in this book was generated specifically for it rather than borrowed from elsewhere, and the assumptions behind each are stated so you can reproduce or challenge them. Where a model rests on assumptions markets violate — and they all do — the book says so. Treat the numbers as illustrations of method, not as predictions.

The callouts

Insights (cyan) are the analytical shifts that change how you interpret your own results. **Warnings** (red) mark the errors that produce confident, wrong conclusions. **Frameworks** (green) are processes you can run this week.

ABOUT

What Drawdown is

Drawdown is a UK trading education platform built on a simple, unfashionable premise: that most people who lose money trading were failed by their education long before they were failed by the market.

The retail space has a content problem. It is saturated with material optimised for engagement rather than accuracy — screenshots without context, strategies without risk parameters, and timelines that would embarrass anyone who has actually done this. The predictable result is a large population of traders who know a dozen chart patterns and have never once calculated whether their results are statistically distinguishable from random.

Drawdown teaches the other part. Risk before entries. Process before profit. Evidence before opinion.

And Signal Centre

Signal Centre (**signalcentre.co.uk**) is a separate platform built for a different audience: an institutional-grade market analysis service for experienced and professional traders, built on multi-factor quantitative analysis rather than opinion.

The distinction between the two businesses is deliberate and worth stating plainly. **Drawdown teaches you to build and measure your own edge.** Signal Centre provides analytical output to traders who already have a process and want an additional institutional-quality input into it. Neither is a substitute for the other, and a trader without their own tested framework will not be rescued by anyone's analysis — a point this book makes at some length.

The standards we hold ourselves to

STANDARD	WHAT IT MEANS IN PRACTICE
No fabricated statistics	Every figure is sourced, computed and reproducible, or clearly labelled illustrative. Nothing is invented to make a point land harder.
No income claims	We will never tell you what you could earn.
Education is never a signal service	Drawdown's education teaches you to build your own process. It does not sell trade alerts as a substitute for skill.
No fake urgency	No countdown timers, no invented scarcity.
Risk stated plainly	The majority of retail accounts lose money. We say so inside the products we ask you to pay for.
Disclosure	Commercial relationships are disclosed at the point of mention.

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ABOUT THE FOUNDER

Who wrote this — and who didn't

My name is Pete Currey. I'm a business builder based in Chesterfield, Derbyshire, and I've spent more than twenty years founding and running companies across facilities management, industrial distribution and digital. I trade my own capital, I hold a funded account with a proprietary trading firm, and I built both Drawdown and Signal Centre.

What I am not is a trading guru, and I'd rather establish that on page eight than have you discover it later. I don't post screenshots of winning trades. I'm not going to tell you what I make, because unverifiable income claims are precisely what makes this industry so hard to trust — and because the number wouldn't help you anyway.

Why this particular book

Because building businesses teaches you something that trading education almost never does: **the difference between a decision that worked and a decision that was correct.** In business, that distinction is everything. Plenty of companies survive bad decisions for years on favourable conditions, then die the moment conditions turn. Plenty of good decisions look wrong for eighteen months.

The only defence, in business and in trading, is measurement — deciding in advance what you're testing, recording honestly what happened, and having the discipline to read the result rather than the outcome you wanted. Most traders never build that apparatus, so they spend years unable to distinguish a working system in drawdown from a broken system in denial. That's the gap this book exists to close, and it's why it contains more statistics than chart patterns.

What this book is built on

The same standard I'd apply to any business I run: every claim is one I'd be comfortable defending, and every figure is one you can reproduce. Where the honest answer is "the model breaks down here" or "this depends on your circumstances and you need a professional", the book says so rather than pretending to an authority it doesn't have.

> WHAT I'M ACTUALLY CLAIMING
>
> That I've built businesses for two decades, that I trade my own money and a funded account, and that I've assembled here the analytical framework I wish had existed when I was stuck — accurate, sequenced, and free of the nonsense.
>
> **That's the whole claim.** Not that I've solved markets. Not that this will make you money. Assess the material on whether it's true and useful, which is the only standard that should ever have applied.

Disclosure: Drawdown holds affiliate relationships with certain companies in this sector, including the firm providing the founder's funded account. No such company is named, recommended or linked anywhere in this book. Signal Centre is a business operated by the author.

The consistency illusion

There is a specific and very common trading story. A trader learns the fundamentals. They find a method that makes sense. They have a good month and conclude they've cracked it. They have a bad month and conclude the method is broken. They find a new method. Repeat for three years.

What makes this story tragic rather than merely unfortunate is that at least one of those discarded methods was probably fine. It was abandoned not because it failed but because **its normal variance was mistaken for failure** — and the trader had no framework capable of telling the difference. They didn't lack discipline or intelligence. They lacked measurement.

The three explanations, and why they're indistinguishable

When your results deteriorate, there are exactly three possible causes:

CAUSE	WHAT'S HAPPENING	CORRECT RESPONSE
Variance	The edge is intact. You're inside the losing stretch every positive-expectancy system produces.	Change nothing. Continue executing.
Decay	The edge has genuinely weakened — conditions changed, participants adapted, the inefficiency closed.	Reduce size, investigate, rebuild or replace.
Drift	The edge is intact but you've stopped executing it — sizing crept, rules loosened, setups degraded.	Return to specification. The strategy needs nothing.

From your equity curve, all three look identical. That is the central problem of this book. Three completely different situations, three opposite correct responses, and one piece of evidence that cannot distinguish between them. Traders guess — and because losing feels like failure, they overwhelmingly guess "the strategy is broken," which is the wrong answer roughly two times out of three.

> DRAWDOWN INSIGHT — THE REFRAME
>
> Consistency is not an outcome you achieve. It is a measurement capability you build. The consistent trader isn't the one whose results don't vary — everyone's results vary. It's the one who can tell you, from data, which of the three explanations applies right now, and therefore what to do. Everything else in this book is in service of that single capability.

Part I diagnoses where you're stuck. Part II gives you the mathematics to interpret your own numbers. Part III covers building and proving an edge properly. Part IV builds the analytics that make the three explanations distinguishable. Part V is about scaling once they are.

Fair warning: this book will probably tell you that your problem is not the one you think it is. That's rather the point.

01

PART I — THE DIAGNOSIS · CHAPTER ONE

What Consistency Actually Means

Why a smooth equity curve is the wrong target, what professionals actually mean by consistency, and the metrics that measure it.

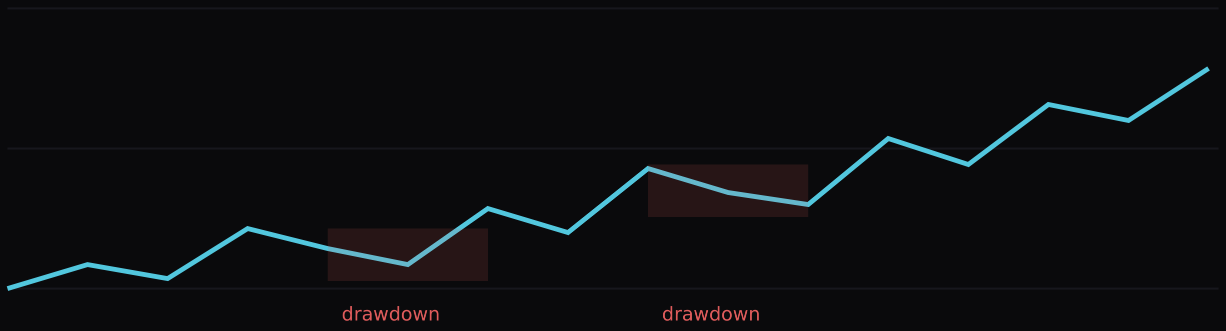
Ask a struggling trader to define consistency and you'll get some version of "making money regularly." Ask a professional and you'll get something closer to "producing the same decisions from the same inputs, and outcomes that fall within the range my system predicted." Those are radically different definitions, and the first one is unachievable by construction.

Why the smooth equity curve is a fantasy

Any strategy with a win rate below 100% produces losing runs. That isn't a flaw to be engineered away; it's an arithmetic certainty. A system winning 40% of the time will, across 200 trades, typically produce a longest losing run of around nine — and one run of fourteen or more in roughly one sample out of twenty. Those are simulated figures for independent trades, and they set a floor on how smooth any equity curve can possibly be.

So when a trader targets "consistent profits every month," they've set an objective their own mathematics forbids. And because they can't hit it, they conclude something is wrong, and they change things — usually the one thing that was working.

A GENUINELY GOOD SYSTEM, HONESTLY PLOTTED



Up over the sample — and unpleasant in at least two places. Every real edge looks like this.

FIG 1.1 — THE EQUITY CURVE OF A PROFITABLE SYSTEM IS NOT A DIAGONAL LINE. IT IS A RISING SEQUENCE INTERRUPTED BY STRETCHES THAT FEEL, AT THE TIME, EXACTLY LIKE FAILURE.

The professional definition

Consistency, properly defined, has three components — and note that none of them is "profit":

COMPONENT	WHAT IT MEANS	HOW IT'S MEASURED
Decision consistency	The same inputs produce the same decision, every time, regardless of recent results	Plan-adherence rate; variance in risk per trade
Statistical consistency	Results fall within the range your system's parameters predict	Rolling expectancy inside its confidence band; drawdowns within simulated range
Operational consistency	The process around trading runs whether or not you feel like it	Review completion rate; journal completeness; data integrity

DRAWDOWN INSIGHT

All three are entirely within your control, and none of them is your P&L. That's not a consolation prize — it's the actual mechanism. Control the three, and your P&L becomes the output of a system whose behaviour you can predict, rather than a number that happens to you.

The variance you must accept

These figures come from simulating 50,000 sequences of 200 independent trades at each win rate. They describe the longest losing run you should expect, not the worst case:

WIN RATE	TYPICAL LONGEST LOSING RUN	1 SAMPLE IN 20 REACHES	1 IN 100 REACHES
60%	5	8	10
50%	7	10	13
45%	8	12	15
40%	9	14	17
35%	11	16	20
30%	13	19	24

Simulated, 200 trades per sequence, 50,000 sequences per win rate, independent outcomes. Real markets cluster — losing runs in practice are often somewhat worse than independence predicts, because market conditions persist.

Read that table against your own experience. If you abandoned a 40%-win-rate system after seven losses in a row, you quit inside the ordinary range — you didn't even reach the typical longest run. That is not a discipline failure but an information failure, and it is extraordinarily common.

> WARNING — THE ASYMMETRY THAT TRAPS EVERYONE
>
> Note the trap in that table: the systems with the best risk-reward profiles have the lowest win rates, and therefore the most brutal losing runs. The strategies most likely to be profitable long-term are precisely the ones most likely to be abandoned — and only knowing the numbers in advance prevents it.

What to measure instead of profit

Replace "am I making money this month" with these four questions, reviewed weekly:

- **Adherence rate.** What percentage of trades followed specification exactly? Target above 95%. Below 85% means you don't currently have a system — you have a set of suggestions.
- **Risk variance.** How much did your risk per trade vary? A standard deviation above 20% of your intended risk means position sizing is being driven by emotion.
- **Rolling expectancy.** Your last 50 trades in R, versus your tested expectancy. Inside the confidence band or outside it? (Chapter 6 shows how to compute the band.)
- **Process completion.** Did the reviews happen? Is the data complete? Missing records are almost always concentrated around the trades you'd least like to examine.

The uncomfortable implication

If those four are healthy and you're still losing, you have a genuine edge problem, and Parts II and III will help. If those four are unhealthy, **no strategy will save you** — because you aren't currently running a strategy, you're running an improvisation that occasionally resembles one. Most traders reading this book will find themselves in the second category, and will find it more useful than they expect.

Chapter 1 — What to take with you

- [ ] A smooth equity curve is mathematically impossible. Stop targeting it.
- [ ] Consistency has three components — decision, statistical, operational. None is profit.
- [ ] Know your system's expected losing run before you experience it.
- [ ] The best risk-reward systems have the worst losing runs. That's why they survive.

02

PART I — THE DIAGNOSIS · CHAPTER TWO

The Six Plateaus

The six reasons competent traders stall — how to identify which one you're on, and why treating the wrong one wastes years.

Traders who plateau nearly always misdiagnose themselves, and the misdiagnosis follows a pattern: they assume the problem is knowledge, because knowledge is the thing that's easiest to buy more of. In practice, competent traders stall for one of six reasons, and only one of them is solved by learning something new.

1 · NO SPECIFICATION
Rules exist in your head, not on paper. Nothing can be tested, so nothing can improve.

2 · NO SAMPLE
You switch methods before any of them accumulates enough trades to mean anything.

3 · EXECUTION GAP
The strategy works. You don't run it. Realised results trail theoretical ones badly.

4 · SIZING FAILURE
Positive expectancy, sized so aggressively that ordinary variance erases the gains.

5 · EDGE DECAY
It genuinely used to work and now doesn't. Conditions moved and you didn't notice.

6 · NO EDGE
Executed perfectly, measured properly, and expectancy is genuinely negative.

THE DIAGNOSTIC ORDER MATTERS
You cannot assess 5 or 6 until 1–4 are resolved. Without specification there is nothing to test; without a sample there is nothing to measure; with an execution gap you are measuring yourself, not your strategy.

FIG 2.1 — THE SIX PLATEAUS. WORK THROUGH THEM IN ORDER. MOST TRADERS SKIP TO 5 AND 6 BECAUSE THOSE BLAME THE MARKET RATHER THAN THE PROCESS.

The order is the most important part of this diagram. Almost every trader who believes they're on plateau 6 ("my strategy doesn't work") is actually on 1, 2 or 3 — and the distinction matters enormously, because the response to plateau 3 is to change nothing about the strategy, while the response to plateau 6 is to abandon it entirely.

Plateau 1 — No specification

Signature: you can describe your approach in conversation but not on paper. Two trades that look similar to you would be classified differently by anyone else. Your entry criteria include words like "strong", "clean" or "obvious".

Why it stalls you: an unwritten strategy cannot be tested, measured, reviewed or improved. Every result is uninterpretable because you can't be sure what produced it. You are, technically, generating random data about a system that doesn't exist.

The fix: Chapter 10. Write the specification until a stranger could apply it and reach your conclusions.

Plateau 2 — No sample

Signature: more than two method changes in the last six months. You can name several strategies you've "tried". Your longest continuous run with one approach is under 40 trades.

Why it stalls you: every switch resets the sample to zero. After three years you have twenty impressions of fifteen systems rather than one properly-tested system — and impressions are worthless, because at low sample sizes results are statistically indistinguishable from random (Chapter 6 quantifies exactly how indistinguishable).

The fix: commit to one specification for a defined minimum number of trades, decided in advance, with no changes permitted inside it regardless of results.

Plateau 3 — Execution gap

Signature: your tested or backtested results are meaningfully better than your live results. Your adherence rate is below 90%. You can point to trades you "should have" taken or "shouldn't have" exited.

Why it stalls you: you're measuring your behaviour and calling it strategy performance. Every conclusion you draw about the strategy is contaminated. Worse, the usual response — changing the strategy — leaves the actual cause untouched, so the cycle repeats indefinitely.

The fix: Chapter 16 quantifies the gap precisely and separates it into its components.

> DRAWDOWN INSIGHT
>
> Plateaus 1–3 are, collectively, where the overwhelming majority of stuck traders actually sit. All three are unglamorous, all three are entirely within your control, and none of them requires learning anything new about markets. That's the good news, though it rarely feels like it — it's much more comfortable to believe you need a better strategy than to accept you haven't been running the one you have.

Plateau 4 — Sizing failure

Signature: profitable in R terms, unprofitable in currency. Your best trades are small and your worst trades are large. Position size varies substantially with confidence.

Why it stalls you: a positive-expectancy system sized inconsistently can produce negative returns, because the losses land disproportionately on the oversized positions. Measuring in R conceals it perfectly — which is why R-multiples must always be checked against currency results.

The fix: Chapter 8 on optimal sizing, plus mechanical enforcement of fixed fractional risk.

Plateau 5 — Edge decay

Signature: a documented period of genuine profitability, followed by sustained deterioration across a large sample. Rolling expectancy trending down over 100+ trades rather than fluctuating. Adherence has stayed high throughout.

Why it stalls you: you're executing correctly on an inefficiency that has closed or a market condition that has changed. Discipline actively hurts here — it keeps you faithfully repeating something that no longer pays.

The fix: Chapter 13. Note the crucial precondition: you can only conclude decay if adherence stayed high. If adherence fell at the same time, you're on plateau 3 wearing plateau 5's clothes, and that's the single most common misdiagnosis in this entire framework.

Plateau 6 — No edge

Signature: a written specification, 200+ trades, adherence above 95%, consistent sizing, and expectancy that is negative or statistically indistinguishable from zero across the full sample.

Why it stalls you: because it's true. The approach doesn't work. This is a legitimate finding, and reaching it properly is genuine progress — you now know something most traders never establish about anything.

The fix: Chapter 9 on where edges come from, and Chapter 21 on deciding whether to rebuild or stop. Note that arriving here honestly requires everything above it to be in place. Very few traders who believe they're on plateau 6 have actually earned the diagnosis.

The diagnostic sequence

FRAMEWORK — RUNNING THE DIAGNOSIS ON YOURSELF

Answer in order. Stop at the first "no" — that's your plateau.

1. Could a stranger read my written rules and identify my setups on a chart without me? No → Plateau 1
2. Do I have 100+ trades of one unchanged specification, recorded? No → Plateau 2
3. Is my adherence rate above 90% across that sample? No → Plateau 3
4. Is my risk per trade consistent, and do my R results and currency results agree? No → Plateau 4
5. Was expectancy previously positive and is it now trending down across 100+ trades with adherence intact? Yes → Plateau 5
6. Everything above is in place and expectancy has never been positive? → Plateau 6

WARNING — THE DIAGNOSIS YOU WANT VS THE ONE YOU HAVE

Nearly everyone wants to be on plateau 5 or 6, because those locate the problem outside themselves. Nearly everyone is on 1, 2 or 3. If your instinct on reading this chapter was to skip to the later plateaus, that instinct is itself diagnostic — and worth recording in your journal.

Chapter 2 — What to take with you

- [ ] Six plateaus, and the order is not optional.
- [ ] Plateaus 1–3 hold most stuck traders and require no new market knowledge.
- [ ] Decay can only be diagnosed if adherence stayed high — otherwise it's drift.
- [ ] Plateau 6 must be earned. Most who claim it haven't done the work to know.

03

PART I — THE DIAGNOSIS · CHAPTER THREE

Auditing Your Own Data

The forensic review of what you have actually been doing — how to reconstruct the record, what to extract from it, and how to read it without flinching.

Before anything can be fixed, you need an honest picture of what has actually happened. Not what you remember happening — memory systematically flatters, rationalising losses, inflating the size of wins, and quietly deleting the trades you'd rather not think about. This chapter is about building a record that can't do that.

The minimum viable dataset

Every trade needs these fields. Anything less and the analysis in Parts II and IV becomes impossible:

FIELD	WHY IT'S NON-NEGOTIABLE
Date & time	Enables session, day-of-week and time-of-day segmentation — often the single most revealing cut of the data.
Instrument	Expectancy frequently differs sharply between instruments you assumed were equivalent.
Setup name	The primary segmentation axis. Without it you can only measure yourself in aggregate, which hides everything.
Planned entry / stop / target	The intended trade. Required to compute the execution gap.
Actual entry / exit	What really happened. The difference between this and the row above is where a great deal of money goes.
Risk in currency	Reveals sizing inconsistency that R-multiples conceal completely.
Result in R	The normalised outcome. All statistical work uses this.
Adherence flag	Binary, ruthless. Did this trade follow specification in every respect — yes or no? Anything less than ruthless makes the field worthless.
Market regime	Trend/range and volatility state at entry. Enables the analysis in Chapter 15 that most traders find the most surprising.

WARNING — THE RECONSTRUCTION TRAP

If you're rebuilding history from broker statements, you can recover entries, exits and sizes — but you cannot recover intent. Adherence, setup name and regime cannot be reconstructed honestly after the fact, because you'll assign them based on the outcome you already know. Mark reconstructed trades as a separate dataset and never mix them with prospectively-recorded ones. Analysing them together produces conclusions that feel rigorous and aren't.

The first five numbers

Before any sophisticated analysis, compute these. They take fifteen minutes and frequently end the investigation early:

1. **Total trades, and trades per specification.** If no single specification has 100+ trades, you're on plateau 2 and the rest of the analysis is premature.
2. **Adherence rate.** Below 90% and you're measuring yourself rather than your strategy.
3. **Mean R and standard deviation of R.** Your raw expectancy and its variability. Both matter; most traders compute only the first.
4. **Risk consistency.** Standard deviation of risk-in-currency divided by mean risk. Above 0.2 indicates emotional sizing.
5. **R result vs currency result.** If R is positive and currency is negative, you have a sizing failure and you'd never have seen it any other way.

The comparison that reveals the most

Split your dataset in two: adherent trades and non-adherent trades. Compute expectancy for each separately. There are three possible results, and each means something very specific:

RESULT	WHAT IT TELLS YOU
Adherent positive, non-adherent negative	The most common outcome by a distance. Your strategy works and your improvisation doesn't. The entire problem is execution, and no strategy change will address it. This finding alone is worth the price of building the dataset.
Both negative	Either the specification is genuinely weak, or your sample is too small to distinguish from noise. Check the sample size requirement in Chapter 6 before concluding anything.
Non-adherent positive	Counterintuitive but not rare, and almost always a small-sample artefact — deviations are infrequent, so their sample is tiny and dominated by one or two outcomes. Do not conclude your instincts beat your rules. Check the sample size, then check it again.

Looking at the tails

Pull your five largest losses and five largest wins and read them individually. Specific questions worth asking: were the large losses adherent trades, or were they the ones where a stop was widened or a size was doubled? Were the large wins the result of the strategy, or of holding beyond target through luck? A profitable record that depends entirely on two outlier wins is not evidence of an edge — it's evidence of two lucky trades, and the strategy underneath may be flat or negative.

DRAWDOWN INSIGHT — THE OUTLIER TEST

Remove your single best trade from the dataset and recompute expectancy. Then remove the best three. If your edge disappears, your edge *is* those trades — and you should be extremely cautious about concluding the system is profitable. Genuine edges survive the removal of their best outcomes with reduced but still positive expectancy.

Auditing what's missing

Gaps in the record are data. Missing entries cluster with astonishing reliability around the trades people least want to examine — meaning your worst behavioural episodes are the ones least represented. Assume the true picture is somewhat worse than the recorded one.

The audit output

FRAMEWORK — THE ONE-PAGE AUDIT SUMMARY

Reduce the entire audit to a single page containing: total trades and trades per specification; adherence rate; expectancy overall, adherent-only and non-adherent-only; standard deviation of R; risk consistency ratio; expectancy with top three trades removed; largest drawdown and its duration; and your plateau diagnosis from Chapter 2.

Date it. Keep it. Repeat quarterly. The change in this page over time is the most honest measure of your development that exists — considerably more honest than your account balance, which is contaminated by luck.

Chapter 3 — What to take with you

- [ ] Memory is not data. Build the record before drawing conclusions.
- [ ] Never mix reconstructed trades with prospectively-recorded ones.
- [ ] Split adherent from non-adherent — that single comparison usually ends the investigation.
- [ ] Remove your top three trades. If the edge vanishes, it was never there.

04

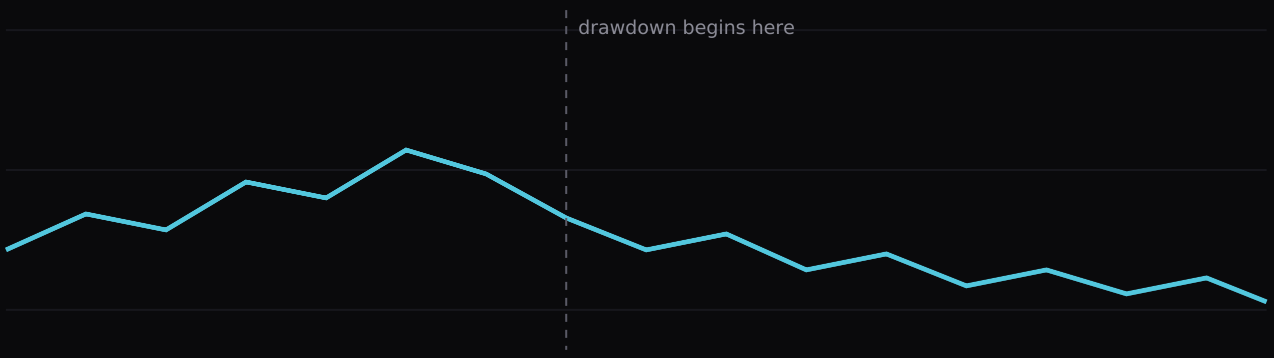
PART I — THE DIAGNOSIS · CHAPTER FOUR

Variance or Decay?

The single most consequential question in trading — and the evidence that actually answers it.

You're in a drawdown. The correct response depends entirely on which of three things is happening, and the equity curve cannot tell you which. Guess wrong in one direction and you abandon a working system at its worst moment. Guess wrong in the other and you keep funding something that has stopped working. This chapter is about not guessing.

Why the equity curve can't answer it



Same curve. Three possible causes: VARIANCE DECAY DRIFT — indistinguishable from this chart alone

FIG 4.1 — EVERY DRAWDOWN LOOKS THE SAME FROM THE OUTSIDE. THE DIAGNOSIS LIVES IN THE UNDERLYING DATA, NEVER IN THE CURVE.

The four tests

Run all four. Together they resolve the question in the large majority of cases.

TEST	WHAT YOU COMPUTE	HOW TO READ IT
1 · Adherence test	Adherence rate during the drawdown vs before it	If adherence fell, you have **drift**. Stop here — this must be resolved before decay can even be assessed.
2 · Magnitude test	Current drawdown vs the simulated distribution of drawdowns for your system (Chapter 7)	Inside the 95th percentile of simulated outcomes → consistent with **variance**. Beyond it → decay becomes more likely.
3 · Composition test	Win rate and average win/loss during the drawdown, separately	Win rate stable but average win shrinking suggests the market's movement has changed. Win rate collapsing with the same average win suggests entry quality has degraded.
4 · Regime test	Segment the drawdown by market regime (Chapter 15)	Losses concentrated in one regime = a regime mismatch, not a broken edge. The system may be fine and simply out of its conditions.

> DRAWDOWN INSIGHT — TEST ONE COMES FIRST FOR A REASON
>
> If adherence declined during the drawdown, every other test is contaminated — you're no longer measuring the strategy, you're measuring a mixture of the strategy and your reaction to losing. This is why "my edge stopped working" is so often wrong: the edge didn't change, the operator did, and the data cannot separate them retrospectively.

The composition test in detail

Expectancy has two moving parts — win rate and the ratio of average win to average loss. When expectancy falls, exactly one of four things has happened, and each points somewhere different:

WHAT CHANGED	LIKELY CAUSE
Win rate down, avg win/loss stable	Entry quality. Either the setup is appearing in worse conditions, or you've loosened your criteria without noticing.
Win rate stable, avg win down	The market is giving less follow-through. Often a volatility regime change — targets that were reachable no longer are.
Win rate stable, avg loss up	Execution. Stops being widened, exits delayed, or slippage increasing. Nearly always behavioural or cost-related, not strategic.
Both deteriorating	Either genuine decay or a substantial regime shift. Test 4 distinguishes them.

This decomposition is remarkably powerful and almost nobody does it. "My results got worse" is unactionable. "My win rate is unchanged but my average win fell from 2.1R to 1.4R while realised volatility halved" points directly at a specific, fixable cause — your targets are calibrated to a market that no longer exists.

How long before you can conclude decay?

Longer than feels reasonable. Chapter 6 works through the arithmetic properly, but the headline is uncomfortable: distinguishing a genuinely deteriorated edge from ordinary variance typically requires a sample of comparable size to the one that established the edge in the first place. If it took 200 trades to demonstrate the edge, expect to need something in that region to demonstrate its decay.

This is why the professional response to suspected decay is not binary. You **reduce size and keep measuring** — protecting capital while the sample accumulates, and preserving the position if the edge is intact. That is a rational price.

FRAMEWORK — THE DRAWDOWN RESPONSE PROTOCOL

1. At -5% or three consecutive losing weeks, trigger the protocol. Do not wait for it to feel serious.
2. Run the adherence test. Adherence down → drift. Fix execution, change nothing else.
3. Adherence intact → halve risk immediately. This is not a signal of no confidence; it's how you buy time to gather evidence.
4. Run tests 2, 3 and 4. Document the findings in writing.
5. Continue at half risk until you have 50+ further trades, then reassess against the tests.
6. Restore full size only when rolling expectancy returns inside its confidence band. Rebuild or retire only on the evidence of tests 2–4, never on how the drawdown feels.

WARNING — THE TWO SYMMETRICAL ERRORS

Abandoning a working system inside normal variance costs you every future pound it would have produced, and it's the more common error by some margin. Persisting with a decayed system costs you capital slowly and continuously. Both are serious. The protocol above is designed to make neither error catastrophic, which is more achievable than trying to always be right.

Chapter 4 — What to take with you

- ☐ Variance, decay and drift look identical on the equity curve. Never diagnose from it.
- ☐ Test adherence first — everything else is contaminated if it fell.
- ☐ Decompose expectancy into win rate and win/loss ratio. The answer is usually there.
- ☐ The response to uncertainty is half size and more data — not a binary decision.

05

PART II — THE MATHEMATICS OF EDGE · CHAPTER FIVE

Expectancy, Properly Understood

The equation everyone quotes and almost nobody uses correctly — its inputs, its failure modes, and what it does not tell you.

Expectancy is the average value of one trade of your system. It is the single number that determines whether repetition helps you or destroys you — and it is routinely computed wrongly, interpreted too confidently, and used to justify decisions it cannot support.

The equation

$$E = (\text{Win\%} \times \text{Average win}) - (\text{Loss\%} \times \text{Average loss})$$

Expressed in R, where 1R is the amount risked per trade.

Positive expectancy means every trade has positive expected value, so trading more — within your risk constraints — increases expected return. Negative expectancy means the reverse: every trade destroys value, and frequency is what kills you rather than direction.

Four systems, one expectancy

These four systems all have roughly the same expectancy. They are not remotely the same to trade:

SYSTEM	WIN RATE	AVG WIN	EXPECTANCY	STD DEV OF R	TYPICAL LONGEST LOSING RUN (200 TRADES)
A	60%	1.0R	+0.20R	0.98	5
B	50%	1.5R	+0.25R	1.25	7
C	40%	2.0R	+0.20R	1.47	9
D	30%	3.0R	+0.20R	1.83	13

All systems assume a 1R loss. Standard deviations computed directly from the outcome distribution; losing runs simulated over 50,000 sequences of 200 independent trades.

Same expected return per trade, radically different experience. System D produces the same money as System A while requiring you to endure losing runs more than twice as long and results that scatter roughly twice as widely. It will also — as Chapter 6 shows — take three to four times as many trades to prove that it works at all.

DRAWDOWN INSIGHT — EXPECTANCY IS NOT ENOUGH

Two systems with identical expectancy are not equivalent investments. You must always quote expectancy alongside its standard deviation, because the second number determines how much capital, patience and evidence you need to realise the first. A trader who knows only their expectancy knows roughly half of what matters.

How expectancy is computed wrongly

- **Mixing specifications.** Aggregate expectancy across three setups tells you nothing useful — one may be strongly positive and two negative. Compute per specification, always.
- **Including non-adherent trades.** That measures you, not the system. Compute both, separately, and compare (Chapter 3).
- **Using currency instead of R** when position sizing varied. Inconsistent sizing makes currency results uninterpretable.
- **Ignoring costs.** Expectancy must be computed net of spread, commission and financing. A gross expectancy of +0.12R can easily be negative after realistic costs, and this is a genuinely common outcome for higher-frequency approaches.
- **Counting scratches as wins.** Break-even exits are 0R, not wins. Classifying them as wins inflates win rate and hides a real problem.

The cost drag, quantified

Costs scale with frequency, and this destroys more strategies than any market condition. Suppose your round-trip cost is 0.1R — a plausible figure when your stop is modest relative to the spread:

GROSS EXPECTANCY	TRADES / YEAR	COST DRAG / YEAR	NET OUTCOME
+0.30R	100	10R	+20R
+0.15R	400	40R	+20R
+0.12R	800	80R	+16R — and far more work
+0.08R	1,000	100R	-20R — profitable gross, losing net

The last row is the one worth sitting with. That trader has a genuine, real, positive edge on every single trade — and loses money, consistently, forever, because they trade it too often at too high a cost. No amount of discipline or psychology fixes that. Only arithmetic does.

Expectancy per unit of time

A refinement most traders never make: expectancy per trade is not the whole picture if your trades occupy wildly different amounts of time and capital. A setup returning +0.4R over four days and one returning +0.2R over two hours are not comparable per trade. Compute **expectancy per opportunity per unit of time** when comparing setups for capital allocation — this is how Chapter 19's risk budgeting decides which setups deserve more of your risk.

What expectancy does not tell you

IT CANNOT TELL YOU	WHY
Whether the number is real	A positive sample expectancy can arise entirely by chance. Chapter 6 quantifies how easily.
Whether it will persist	Expectancy is a historical measurement. Edges decay (Chapter 13).
What drawdown you'll endure	That depends on the distribution and sequence, not the mean (Chapter 7).
How much to risk	That requires expectancy and variance and your tolerance (Chapter 8).
Whether outliers carry it	A mean is easily dominated by two trades. Run the outlier test from Chapter 3.

WARNING — THE MOST SEDUCTIVE ERROR IN TRADING ANALYTICS

Computing a positive expectancy from 30 trades and treating it as established fact. It is one of the most common causes of overconfident sizing, and it is precisely how competent traders blow accounts. The number is real; the confidence is not. Chapter 6 exists entirely because of this error.

Chapter 5 — What to take with you

- [ ] Always quote expectancy with its standard deviation. One without the other misleads.
- [ ] Compute per specification, net of costs, adherent trades only.
- [ ] A genuine positive edge traded too frequently at too high a cost is a losing system.
- [ ] Expectancy measures the past. It says nothing about certainty or persistence.

06

PART II — THE MATHEMATICS OF EDGE · CHAPTER SIX

Sample Size & the Noise Floor

How many trades before your results mean anything — and why the honest answer is far larger than anyone told you.

This is the most important chapter in the book, and the one most likely to irritate you. It answers a question almost no retail trader ever asks: given the results I have, how confident can I actually be that my edge is real rather than luck? The answer is usually "far less than you assumed," and understanding why changes how you make every subsequent decision.

The noise floor

Every set of results is a mixture of signal (your edge) and noise (random variation). Below a certain sample size the noise dominates so completely that the results are indistinguishable from random. That's the **noise floor**, and results beneath it carry no information whatsoever — not weak information, none.

A concrete demonstration. Take a system with genuinely zero edge — a coin flip with 1R risk and 1R reward. Simulate many sequences of 30 trades. A substantial proportion of those sequences will show a comfortably positive result purely by chance. If a trader ran one of those sequences, they would conclude they had an edge, size up accordingly, and be entirely wrong. Nothing in their data would have told them otherwise.

The arithmetic of confidence

The standard error of your measured expectancy tells you how much it would be expected to vary from sample to sample:

$$\text{Standard error} = \sigma \div \sqrt{n}$$

$$95\%\ \text{confidence interval} \approx \text{measured expectancy} \pm 1.96 \times \text{SE}$$

Where σ is the standard deviation of your trade results in R, and n is the number of trades. For your edge to be statistically distinguishable from zero at the 95% level, your measured expectancy must exceed 1.96 standard errors. Rearranging gives the number of trades required:

$$n \approx (1.96 \times \sigma \div E)^2$$

What that actually means in practice

SYSTEM	WIN RATE	AVG WIN	EXPECTANCY	Σ	TRADES NEEDED FOR 95% CONFIDENCE
A	60%	1.0R	+0.20R	0.98	≈ 92
B	50%	1.5R	+0.25R	1.25	≈ 96
C	40%	2.0R	+0.20R	1.47	≈ 207
D	30%	3.0R	+0.20R	1.83	≈ 323
E	45%	1.5R	+0.125R	1.24	≈ 380
F	35%	2.0R	+0.05R	1.43	≈ 3,146

Computed directly from each system's outcome distribution, assuming 1R losses and independent trades. Real-world requirements are typically higher, because market outcomes are not fully independent.

Sit with system F. A real, positive edge of +0.05R per trade — and over three thousand trades needed to establish that it exists. At twenty trades a week that's three years. Most traders with a genuinely small edge will never accumulate enough evidence to know it.

> DRAWDOWN INSIGHT — THE THREE CONSEQUENCES
>
> **First:** your 30-trade result means nothing, whatever it shows. **Second:** low win rate systems need far more evidence, so trend-following approaches take dramatically longer to validate than mean-reversion ones. **Third, and most usefully:** a large edge is easier to prove than a small one — which is an argument for pursuing fewer, higher-quality opportunities rather than optimising a marginal system you'll never be able to verify.

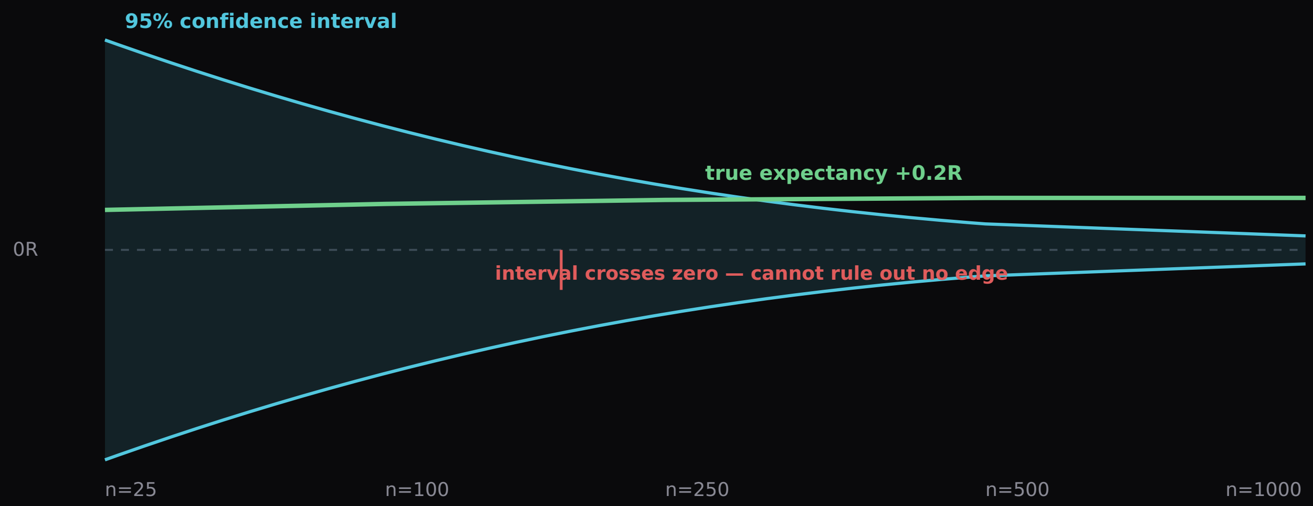


FIG 6.1 — THE CONFIDENCE INTERVAL NARROWS WITH THE SQUARE ROOT OF SAMPLE SIZE. UNTIL IT STOPS CROSSING ZERO, "I HAVE AN EDGE" IS NOT A SUPPORTED CLAIM.

The square root problem

Precision improves with **√n**, not n. Halving your uncertainty needs four times the trades. This is why patience is a mathematical requirement rather than a virtue — and why switching systems every forty trades guarantees permanent uncertainty about everything you've done.

Practical thresholds

SAMPLE	WHAT YOU MAY LEGITIMATELY CONCLUDE
< 30	Nothing. Not "early signs", not "promising". Nothing.
30–100	Whether the specification is executable in practice, and roughly what its win rate and R-distribution look like. Not whether it's profitable.
100–200	A directional indication for high-expectancy, low-variance systems only. Compute your own interval.
200–500	Reasonable confidence for most retail systems, provided expectancy is 0.2R or better.
500+	Confidence in the edge, and enough data to segment by setup, session and regime (Chapter 15).

> WARNING — MULTIPLE COMPARISONS
>
> If you test twenty variations of a strategy and one shows a significant result at the 95% level, you have not found an edge — at that threshold, roughly one in twenty tests produces a false positive by chance alone. This is precisely how curve-fitted backtests are born. Decide what you're testing before you look, and treat anything discovered by searching as a hypothesis requiring fresh out-of-sample data (Chapter 11).

A worked example — reading your own numbers

Suppose you have 140 trades of one specification. Mean result +0.18R, standard deviation 1.35R. Compute the standard error: $1.35 \div \sqrt{140} = 1.35 \div 11.83 =$ `0.114`. The 95% interval is therefore $0.18 \pm (1.96 \times 0.114)$, which is `-0.04R to +0.40R`.

That interval includes zero. Your best estimate is a healthy +0.18R, and yet the evidence is genuinely consistent with having no edge at all. This does not mean you should stop — it means you should not size as though the edge is established, and you should not draw structural conclusions from any subset of these 140 trades. Required sample for confidence here: $(1.96 \times 1.35 \div 0.18)^2 \approx$ **216 trades**. You are two-thirds of the way there.

Now the same calculation after a bad month takes you to 170 trades and a mean of +0.11R. Standard error 0.104, interval -0.09R to +0.31R. Nothing has been proven or disproven; you simply still don't know. The correct action is unchanged: keep executing, keep recording, don't touch the specification. Being able to say that with a number behind it — rather than as reassurance — is precisely the capability this chapter exists to build.

Chapter 6 — What to take with you

- [ ] Compute your own confidence interval: $E \pm 1.96 \times \sigma/\sqrt{n}$. If it crosses zero, you don't know.
- [ ] Precision improves with $\sqrt{n}$. Four times the trades to halve the uncertainty.
- [ ] Low win rate systems require far more evidence. Budget for it before you start.
- [ ] Testing many variations manufactures false positives. Decide before you look.

07

PART II — THE MATHEMATICS OF EDGE · CHAPTER SEVEN

Thinking in Distributions

Your results are not a line, they are a range of possible worlds — and the one you're living in was substantially decided by sequence.

Your equity curve is one draw from a distribution of curves your system could have produced. Change nothing but the order of the same trades and you get a different curve, a different maximum drawdown, and quite possibly a different decision about whether to continue. Understanding that properly is what separates evaluating a system from reacting to an outcome.

Sequence risk

Take 200 trades with a fixed set of outcomes. Their sum is identical regardless of order. But the path is not: if the losses cluster early you may experience a drawdown that ends the account, or your confidence, before the wins arrive. Same trades, same edge, same expectancy — different outcome entirely, decided by nothing but sequence.

This is why "it works over the long run" is inadequate as a defence. You must survive the short run to reach the long run, and the short run is where sequence lives.

Monte Carlo simulation

The tool that makes this tractable. You take your system's parameters, generate thousands of possible sequences, and examine the resulting distribution rather than any single path. It requires no specialist software — a spreadsheet handles it, and Chapter 17 sets out how.

The tables below simulate **System C** (40% win rate, +2R wins, -1R losses, expectancy +0.20R) over 200 trades, 20,000 sequences, compounding, at four different risk levels:

RISK / TRADE	WORST 5% END AT	MEDIAN END	BEST 5% END AT	TYPICAL MAX DRAWDOWN	MAX DD IN WORST 5%
0.5%	1.03×	1.21×	1.43×	5.9%	11.0%
1.0%	1.05×	1.46×	2.03×	11.9%	21.0%
2.0%	1.06×	2.04×	4.16×	23.1%	38.3%
3.0%	0.94×	2.74×	7.93×	33.0%	52.1%

Simulated: 20,000 sequences of 200 independent trades, compounding, no costs. Assumes independence and stationarity — both of which real markets violate, so treat these as a floor on the range of outcomes rather than a ceiling.

Several things in that table deserve attention. Note that the **median** outcome improves as risk rises — which is exactly why aggressive sizing is tempting. Note also that the **worst 5%** barely improves at all, and at 3% risk actually turns negative: the same positive edge, and one sequence in twenty leaves you down after 200 trades. And note that maximum drawdown scales almost linearly with risk while the median gain does not.

> DRAWDOWN INSIGHT — WHAT YOU'RE REALLY CHOOSING
>
> Increasing risk buys you a better median outcome at the cost of a dramatically worse bad case, and the bad case is the one that ends careers. At 0.5% risk this system has essentially no path to a serious drawdown; at 3% it has a one-in-twenty path to a 52% drawdown, from which recovery requires more than doubling. Same edge, same trades. The only variable is what you chose to survive.

The drawdown you should expect

Most traders have no idea what drawdown their system should produce, so they treat every drawdown as evidence of failure. Simulation replaces that with a number. If the median maximum drawdown for your system at your risk level is 12%, then a 10% drawdown is entirely ordinary and requires no action whatsoever — a fact worth knowing in advance rather than discovering emotionally.

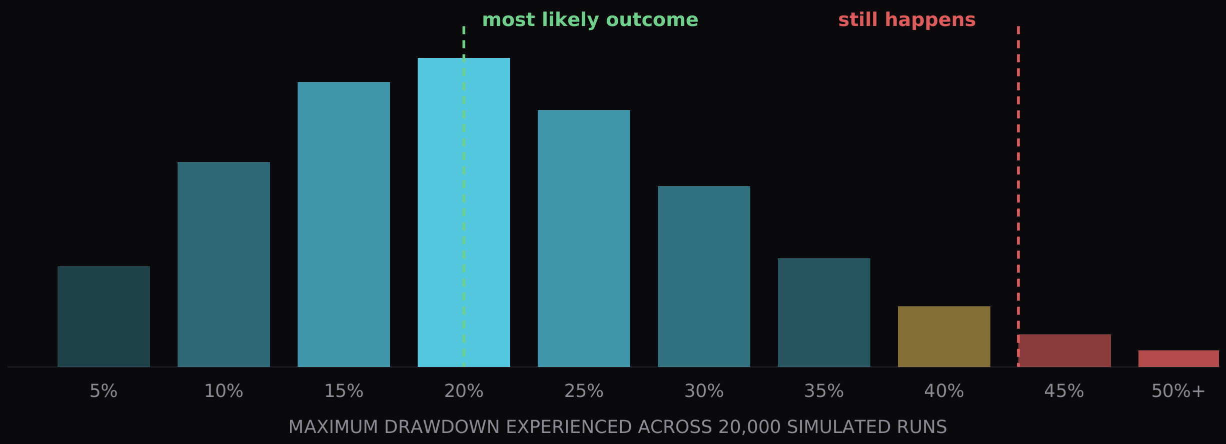


FIG 7.1 — MAXIMUM DRAWDOWN IS A DISTRIBUTION, NOT A NUMBER. THE TAIL IS THIN BUT IT IS NOT EMPTY, AND YOU MUST BE SIZED TO SURVIVE IT.

Reshuffling your own trades

You don't need theoretical parameters — you can bootstrap from your own history. Take your recorded R-results, shuffle them randomly, rebuild the equity curve, and repeat a few thousand times. This tells you what your actual system could plausibly have produced, and it is a genuinely uncomfortable exercise: most traders discover that their real result sits comfortably inside a range that includes considerably worse outcomes they simply didn't happen to draw.

FRAMEWORK — THE BOOTSTRAP IN A SPREADSHEET

1. List your R-results in a column. **2.** In a new column, sample from them at random with replacement, as many rows as your sample size. **3.** Build a running equity curve from that column, compounding at your actual risk. **4.** Record final equity and maximum drawdown. **5.** Recalculate a few thousand times and collect the results. **6.** Read the 5th, 50th and 95th percentiles of both final equity and maximum drawdown.

Those six numbers tell you more about your system than a year of watching your equity curve.

WARNING — THE ASSUMPTIONS THIS RESTS ON

Bootstrapping assumes trades are independent and that the distribution is stable. Neither is strictly true: market conditions persist, so losses cluster more than random reshuffling implies, and distributions shift over time. Real drawdowns therefore tend to be somewhat worse than simulation suggests. Treat simulated figures as an optimistic floor, not a forecast.

Chapter 7 — What to take with you

- [ ] Your equity curve is one draw from a distribution. Judge the distribution, not the draw.
- [ ] Know your expected drawdown before you experience it.
- [ ] Higher risk improves the median and devastates the bad case. Choose accordingly.
- [ ] Bootstrap your own results. Six percentile numbers beat a year of watching.

08

PART II — THE MATHEMATICS OF EDGE · CHAPTER EIGHT

Risk of Ruin & Optimal Sizing

The Kelly criterion, why almost nobody should use it in full, and how professionals actually decide how much to risk.

Given a known edge, there is a mathematically optimal amount to risk per trade. Understanding that number is valuable. Trading it is close to insane. This chapter explains both halves of that statement, and what to do instead.

Risk of ruin

Risk of ruin is the probability that variance exhausts your capital before your edge pays you. It rises with risk per trade, falls with edge size, and rises sharply with the variance of your outcomes. Its critical property is that it is **non-linear**: doubling your risk per trade does far more than double your probability of ruin.

This is the central asymmetry of position sizing. Gains from higher risk are roughly proportional; the danger is exponential. That relationship is why professional risk management looks so conservative to outsiders — it isn't timidity, it's an accurate reading of a curve.

The Kelly criterion

Kelly gives the fraction of capital that maximises the long-run geometric growth rate:

$$f^* = p - q \div b$$

p = win probability · q = 1 - p · b = ratio of average win to average loss

SYSTEM	WIN RATE	AVG WIN	EXPECTANCY	FULL KELLY	QUARTER KELLY
A	60%	1.0R	+0.20R	20.0%	5.0%
B	50%	1.5R	+0.25R	16.7%	4.2%
C	40%	2.0R	+0.20R	10.0%	2.5%
D	30%	3.0R	+0.20R	6.7%	1.7%

Look at System A: full Kelly says risk **20% of your capital per trade**. Follow that and a run of five losses — which this system produces regularly — leaves you down roughly two-thirds. The mathematics is correct; the recommendation is unusable.

WARNING — WHY FULL KELLY DESTROYS REAL TRADERS

Kelly assumes you know p and b **exactly**, that they never change, that outcomes are independent, and that you can tolerate drawdowns exceeding 50% without deviating. Every one of those assumptions fails in trading. Worse, Kelly is acutely sensitive to input error: overestimate your edge — which, per Chapter 6, you almost certainly have — and Kelly sizing moves rapidly from optimal to ruinous. Overestimating edge by a factor of two produces sizing that can turn a profitable system into a losing one.

Why the curve is asymmetric

FIG 8.1 — THE GROWTH CURVE IS FLAT NEAR ITS PEAK AND FALLS AWAY STEEPLY BEYOND IT. BETTING WELL BELOW OPTIMAL COSTS YOU LITTLE; BETTING ABOVE IT COSTS YOU EVERYTHING.

This shape is the practical justification for conservative sizing. Because the curve is nearly flat around its maximum, a quarter of Kelly captures most of the available growth. Because it collapses beyond the peak, a modest overestimate of your edge — combined with sizing near Kelly — produces negative growth from a positive-expectancy system. Given that you cannot know your true edge precisely, deliberately sizing well below the estimated optimum is not caution. It's the correct response to uncertainty.

How professionals actually decide

FRAMEWORK — SIZING FROM FOUR CONSTRAINTS

Take the **smallest** figure the four constraints permit. Not the average — the smallest.

1. Statistical constraint. Compute Kelly from your measured parameters, then divide by four or more. Divide by more if your sample is under 200 trades.
2. Drawdown constraint. Simulate (Chapter 7). Find the risk level at which your 95th-percentile maximum drawdown stays within what you can tolerate both financially and emotionally.
3. Structural constraint. Any hard external limit — a prop firm's maximum drawdown, a personal capital floor, a business requirement.
4. Behavioural constraint. The size at which you reliably follow your plan. If you interfere with positions above 0.5%, your optimal risk is 0.5%, whatever the mathematics says.

For most retail traders these four converge on something between 0.25% and 1% per trade — which is exactly what conventional advice recommends. The difference is that you now know why, and can defend it, rather than following a rule of thumb you'll abandon under pressure.

Sizing under uncertainty

Confidence should scale risk. Size fully only when your sample exceeds the Chapter 6 requirement and rolling expectancy sits inside its band; half when between half and full requirement; minimum below that. Uncertainty becomes an explicit input rather than something resolved by feel.

Chapter 8 — What to take with you

- [ ] Risk of ruin rises non-linearly. Gains are proportional; danger is exponential.
- [ ] Kelly is a useful upper bound and an unusable instruction. Quarter it, at least.
- [ ] The growth curve is flat below the peak and collapses above it. Err low deliberately.
- [ ] Take the smallest of four constraints — statistical, drawdown, structural, behavioural.

09

PART III — BUILDING & PROVING AN EDGE · CHAPTER NINE

Where Edges Actually Come From

The four sources of genuine advantage, which are realistically available to you, and why the most accessible one is the one nobody wants.

An edge is a persistent reason why your expected return is positive. If you cannot articulate that reason — not the pattern, the reason — then what you have is a correlation you found in historical data, and correlations found by searching disappear at roughly the rate you'd expect from chance.

The four sources

SOURCE	WHAT IT IS	AVAILABILITY TO YOU
Informational	Knowing something relevant before the market prices it	**Effectively none.** Institutions spend enormous sums here and some of it is illegal for you. Discard this category entirely.
Structural	Persistent features of how markets operate — session behaviour, liquidity patterns, participants who must trade for non-profit reasons	**Real but modest.** Requires genuine study of one instrument over a long period. This is where most legitimate retail edges live.
Behavioural	Other participants' predictable errors — stops clustered at obvious levels, herding, capitulation, overreaction to news	**Real and durable.** Human psychology doesn't update. This is the most reliable source available to retail.
Executional	Realising more of a known edge than others do — better adherence, smaller average losses, lower costs, no missed trades	**Fully available, and largely ignored.** Chapter 16 shows this is frequently worth more than any strategy change.

DRAWDOWN INSIGHT — THE EDGE NOBODY WANTS

The executional edge is the only one you can improve unilaterally, this week, with no new market knowledge. If your realised results trail your specification's theoretical results by 0.1R per trade — a very ordinary gap — then closing it is worth exactly as much as finding a whole new strategy with 0.1R expectancy. One takes months of research and might not exist. The other requires you to follow rules you've already written.

Why edges exist at all

An edge requires someone on the other side accepting a worse expected outcome. There are only a few honest reasons that happens: they're trading for a non-profit purpose (hedging, rebalancing, meeting a mandate); they're constrained (must trade at a particular time or size); they're making a predictable behavioural error; or they're being compensated for taking a risk you're accepting instead.

If you cannot name which of those applies to your setup, be sceptical of it. This single question kills more bad strategies than any backtest, and it costs nothing to ask. "It works because the 50 EMA is magic" is not an answer. "It works because stops cluster above obvious highs and larger participants access that liquidity" is — and it's testable.

Behavioural edges in practice

These persist because they're rooted in human wiring rather than in any exploitable inefficiency that arbitrage can close. The predictable patterns: traders place stops in obvious places; losses are cut late and gains taken early; recent information is overweighted; crowds capitulate together at extremes; and round numbers attract disproportionate order flow. None of these is a strategy on its own — but each is a genuine reason, which is what a specification needs underneath it.

Structural edges in practice

These come from market mechanics: sessions with reliably different characteristics, liquidity thinning at particular times, behaviour around scheduled events. They require knowing one instrument deeply rather than many superficially — which is why most retail traders never develop one.

Edge stacking

Individual retail edges are small. The practical route to a workable system is combining several small, independent advantages:

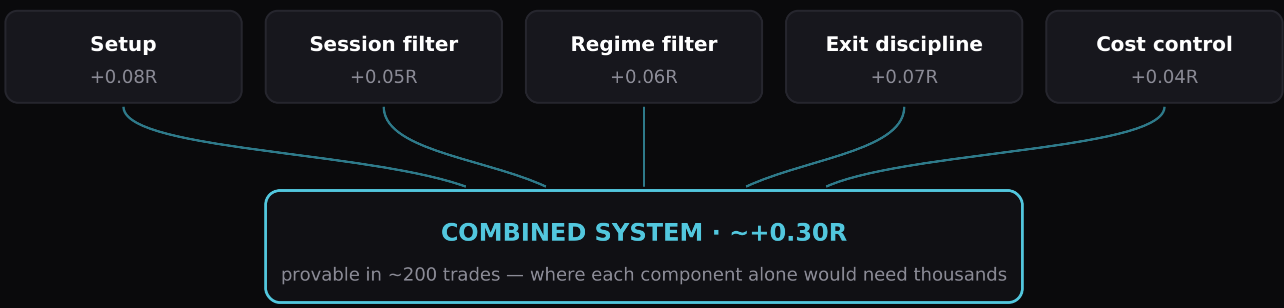


FIG 9.1 — ILLUSTRATIVE. SMALL INDEPENDENT ADVANTAGES COMBINE INTO A SYSTEM LARGE ENOUGH TO BE BOTH TRADEABLE AND, CRUCIALLY, PROVABLE WITHIN A REALISTIC NUMBER OF TRADES.

Note the second line of that diagram carefully, because it's the point most traders miss. A +0.05R edge requires thousands of trades to verify (Chapter 6). A +0.30R edge requires a couple of hundred. **Stacking isn't just about making more money — it's about building something you can actually prove exists within your lifetime as a trader.**

WARNING — INDEPENDENCE IS THE WHOLE GAME

Components only stack if they're genuinely independent. Three filters that all measure momentum are one filter counted three times, and combining them adds complexity, costs degrees of freedom in testing, and delivers no additional edge. Before adding a component, ask what distinct question it answers that nothing else in the system does.

Testing whether a reason is real

FRAMEWORK — THE FOUR-QUESTION EDGE TEST

1. Who is on the other side, and why are they accepting a worse outcome? If you can't answer, stop here.
2. Why hasn't this been arbitraged away? Acceptable answers: it's behavioural and humans don't update; it's too small for institutions; it requires discretion that doesn't scale.
3. What would have to change for this to stop working? If you can't name it, you can't monitor for it (Chapter 13).
4. Does it survive out-of-sample? The reason must hold in data you didn't use to find it (Chapter 11).

Chapter 9 — What to take with you

- ☐ An edge is a reason, not a pattern. Name the reason or discard the setup.
- ☐ Informational edges are unavailable. Behavioural and structural ones are real but small.
- ☐ The executional edge is free, immediate, and the one most people ignore.
- ☐ Stack independent components — it makes the edge both larger and provable.

10

PART III — BUILDING & PROVING AN EDGE · CHAPTER TEN

Specification

Turning a vague idea into a rule set precise enough to be tested, measured and improved — and the discipline of leaving nothing to judgement.

A specification is a document that would allow a competent stranger to produce the same trades you would, from the same charts, without asking you a single question. Anything less is not a strategy — it's a habit with a name, and habits cannot be tested.

The specification test

Write your rules. Hand them to someone else along with a chart. Have them mark every setup they find. Compare with your own marks. **Every disagreement is an unwritten rule** — a piece of judgement you're applying without having documented it. Write it down and repeat. You are finished when the disagreement rate approaches zero.

Most traders discover on first attempt that their disagreement rate is enormous, which is itself the finding: they have been running a different strategy every week without realising, because the undocumented parts drift with mood, recent results and market conditions.

The components, in required detail

COMPONENT	WHAT "SPECIFIED" ACTUALLY MEANS
Universe	Exact instruments. Not "major FX pairs" — the specific pairs, and why those.
Timeframes	Context, decision and execution timeframes named explicitly.
Regime filter	The observable, measurable condition that must hold. "Trending" is not measurable; "price above the 200-period average on the daily and the last three swing lows ascending" is.
Setup conditions	An ordered checklist where every item is binary. If an item requires judgement, decompose it until it doesn't.
Entry trigger	The precise, observable event. Including how long the trigger remains valid if you're not watching.
Stop placement	A rule, not a distance. "Below the swing low that formed the setup, plus a buffer of 0.2× ATR."
Target logic	Fixed R, structural level, trailing rule, or time-based exit — and what happens if none is reached.
Management	Break-even rules, partial exits, and their exact triggers. If you have none, write "none" — that is also a specification.
Exclusions	Conditions under which a valid setup is skipped: news windows, holidays, post-daily-stop, low liquidity.
Sizing	The formula, and the conditions under which it changes.

> DRAWDOWN INSIGHT — DEGREES OF FREEDOM
>
> Every parameter you add is a dial you can turn, and every dial multiplies the number of combinations you could test. Test enough combinations and one will look excellent by chance alone — this is exactly the multiple-comparisons trap from Chapter 6. A specification with four parameters is far more likely to survive out-of-sample than one with fifteen. **Simplicity isn't aesthetic preference here; it's statistical protection.**

Making judgement explicit

Some approaches genuinely involve discretion. That's permissible — but the discretion must be bounded and recorded. If you allow yourself to skip a valid setup on judgement, then you must log every skip with its reason, because those skips are part of your strategy whether you acknowledge them or not.

This matters more than it sounds. A trader who takes 60% of their valid signals is running a different system from their specification, and their measured expectancy applies to the wrong thing entirely. Log the skips, then analyse them: if skipped trades would have been profitable on average, your discretion is costing you money and should be removed. If they'd have lost, your discretion is an edge component and should be written into the specification properly.

Versioning

Treat your specification like software. Give it a version number and a date. When you change it, increment the version and record what changed, why, and what evidence prompted it. Then **start a fresh sample** — results from version 1.2 do not belong in the same dataset as version 1.3.

This sounds fussy until you realise the alternative: most struggling traders cannot tell you what their strategy was six months ago, so they cannot tell whether their results improved because of a change or in spite of it. Versioning converts vague evolution into a series of testable experiments.

WARNING — THE SILENT REVISION

The most damaging changes are the ones you don't notice making: a criterion you start interpreting loosely, a filter you stop checking, a target you begin taking early. These never get a version number, so your dataset silently becomes a mixture of several systems. This is the mechanism behind an enormous proportion of "my edge stopped working" — the specification didn't decay, it quietly mutated.

The specification document

FRAMEWORK — WHAT YOUR DOCUMENT CONTAINS

Header: name, version, date, author.
Rationale: the answers to the four-question edge test from Chapter 9. Why does this work, who's on the other side, what would break it.
Rules: all ten components above, each unambiguous.
Parameters: every number in one place, so you can see how many dials exist.
Test record: in-sample results, out-of-sample results, forward-test results, live results — each with sample size and date range.
Monitoring: the metrics you'll watch and the thresholds that trigger review (Chapter 13).
Change log: every version, what changed, and the evidence that prompted it.

Two to four pages. If it's longer, you probably have too many parameters. If it's shorter than a page, it isn't finished.

Chapter 10 — What to take with you

- [ ] The test: a stranger reproduces your trades from the document alone.
- [ ] Every disagreement reveals an unwritten rule. Write it and repeat.
- [ ] Fewer parameters survive out-of-sample. Simplicity is statistical protection.
- [ ] Version everything. Silent revisions are how edges appear to decay.

11

PART III — BUILDING & PROVING AN EDGE · CHAPTER ELEVEN

Backtesting Without Fooling Yourself

The biases that make historical testing lie to you, and the out-of-sample discipline that is the only real defence.

Backtesting is the cheapest way to gather evidence and the easiest way to generate false confidence. Nearly every backtest a retail trader runs is optimistic, often dramatically so, and the traders least aware of this are the ones who spend the most time optimising. This chapter is about testing in a way that can actually tell you you're wrong.

The seven biases

BIAS	HOW IT INFLATES YOUR RESULTS	DEFENCE
Hindsight	Seeing what happened next makes entries obvious that weren't	Cover the right of the screen. Step forward candle by candle, never scroll back.
Curve fitting	Tuning parameters until they fit history perfectly — fitting noise, not signal	Minimise parameters; test only round, sensible values; validate out-of-sample.
Selection	Recording the clean examples and skipping the ambiguous ones	Log every signal the rules generate, including the ones you dislike.
Survivorship	Testing on instruments or periods that still exist and still trend	Include difficult regimes deliberately — 2008, 2020, extended ranges.
Look-ahead	Using information not available at decision time — a close price to trigger an intrabar entry, a revised data figure	Audit every rule for exactly when its inputs became knowable.
Cost omission	Ignoring spread, commission, financing and slippage	Apply realistic costs — and stress them at double.
Optimisation	Testing many variants and reporting the best one	Count how many variants you tried. That number determines how sceptical to be.

WARNING — THE SINGLE MOST COMMON FAILURE

Testing on all your data, tuning until results look good, then trading it. This is guaranteed to produce an optimistic result and provides no information about future performance — because you have not tested the strategy, you have described the history. The out-of-sample split below is not an optional refinement. Without it, a backtest is decoration.

The out-of-sample split

IN-SAMPLE — 60%	VALIDATION — 20%	LOCKED — 20%
develop and tune here, freely	check, once	touch once, ever

Performance degradation from in-sample to out-of-sample is normal and expected. A drop of roughly a third is unremarkable. A drop to zero or negative means you fitted noise. If you return to the locked segment after a failure and adjust, it is no longer out-of-sample — it has become in-sample, permanently.

FIG 11.1 — THE DISCIPLINE IS SIMPLE TO DESCRIBE AND DIFFICULT TO FOLLOW. ITS ENTIRE VALUE LIES IN THE FACT THAT YOU ONLY GET TO LOOK ONCE.

Walk-forward testing

A more rigorous approach that better simulates reality. Rather than one split, you roll through history: optimise on a window, test on the period immediately following, then move both windows forward and repeat. The aggregate of all the out-of-sample periods is your walk-forward result.

Its advantage is that it mimics what you'd actually do — periodically reassessing on recent data and trading forward. Its finding is often deflating: strategies that look excellent on a single optimised backtest frequently produce mediocre walk-forward results, because each re-optimisation was fitting to conditions that then changed. That's not a flaw in the method; it's the method working.

Manual versus automated

APPROACH	STRENGTHS	WEAKNESSES
Manual	Builds genuine market intuition; forces you to confront ambiguity in your own rules; catches specification gaps immediately	Slow; small samples; vulnerable to hindsight unless disciplined
Automated	Large samples; perfectly consistent rule application; fast iteration	Requires fully mechanical rules; makes over-optimisation trivially easy; hides data quality problems

For a discretionary trader, manual first is almost always correct — hand-testing 100 trades exposes more holes in a specification than any automated run.

Stress testing

FRAMEWORK — FOUR STRESSES EVERY BACKTEST SHOULD SURVIVE

1. Cost stress. Double your assumed costs. Does the edge survive? If it dies, it was a cost-sensitive strategy and live trading will be worse than your assumption.
2. Slippage stress. Assume every entry fills 10% of your stop distance worse and every stop fills 10% beyond. Recompute.
3. Missed-trade stress. Randomly remove 20% of trades — you will miss some. Does the result hold, or was it carried by a handful of outcomes?
4. Parameter stress. Shift each parameter by ±20%. Results should degrade gracefully. If a small change destroys the edge, you've fitted noise and the parameter value was never real.

Parameter stress is the most revealing of the four. A genuine edge sits on a plateau of similar values; a curve-fitted one sits on a spike.

What a good backtest result looks like

Modest. Positive expectancy that survives out-of-sample with reduced but intact performance, degrades gracefully under all four stresses, doesn't depend on a small number of outlier trades, and produces a drawdown profile you could actually tolerate. A backtest showing spectacular returns is not good news — it is nearly always evidence of a methodological error, and the appropriate first response is to go looking for it.

Chapter 11 — What to take with you

- [ ] A backtest without an out-of-sample split is decoration, not evidence.
- [ ] You get to look at locked data once. After that it isn't out-of-sample.
- [ ] Stress costs, slippage, missed trades and parameters. Real edges degrade gracefully.
- [ ] Spectacular results indicate a mistake. Go and find it.

12

PART III — BUILDING & PROVING AN EDGE · CHAPTER TWELVE

Forward Testing & the Live Gap

Why tested results and live results diverge, how much divergence is normal, and how to run the transition so the difference is measurable.

Every strategy performs worse live than in testing. This is not a failure — it is a structural certainty with identifiable causes, most of which can be measured and some of which can be reduced. What matters is knowing how large the gap should be, so you can tell an ordinary transition from a broken one.

The five causes of the live gap

CAUSE	TYPICAL MAGNITUDE	REDUCIBLE?
Execution slippage	Entries and exits fill worse than assumed, especially in fast conditions	Partly — via order type, timing and instrument choice
Missed trades	You weren't watching, hesitated, or the setup formed while you were away	Substantially — via alerts, pending orders, session discipline
Ambiguity resolution	In testing, borderline setups get resolved calmly; live, they get resolved by mood	Substantially — via better specification (Chapter 10)
Behavioural interference	Early exits, widened stops, skipped signals after losses	Yes — and this is usually the largest single component
Regime difference	The live period simply isn't like the tested period	No — but it is detectable via segmentation (Chapter 15)

DRAWDOWN INSIGHT — THE GAP IS DATA, NOT DISAPPOINTMENT

Most traders treat underperformance versus backtest as evidence the backtest was wrong. Often it is — but just as often the strategy is fine and the gap is entirely explained by the five causes above, four of which are yours. Measuring the gap by component tells you which. That measurement is Chapter 16, and it's the most commercially valuable analysis in this book.

The forward test protocol

FRAMEWORK — PROVING A STRATEGY FORWARD

Stage 1 — Paper, real time. Minimum 50 trades. Record intended entry, stop, target and size before the outcome. This tests whether the specification is followable in real time, which backtesting cannot.

Stage 2 — Live, minimum size. Minimum 50 trades. The purpose is not profit; at this size it's irrelevant. The purpose is to measure the behavioural component of the gap under real money conditions.

Stage 3 — Live, half size. Minimum 50 trades. Adherence must hold above 95%. If it deteriorates, return to Stage 2 — the constraint is behavioural capacity, not confidence.

Stage 4 — Live, full calculated size. Only once the sample requirement from Chapter 6 is met and the gap has been measured and is stable.

Roughly 200 trades before full size. At twenty trades a week that's ten weeks; at five a week it's most of a year. Both are correct — the number of trades matters, the calendar does not.

Recording intent

The single most important discipline in forward testing: record what you intended before you know what happened. Planned entry, planned stop, planned target, planned size — written down at decision time. Without this, the execution gap is unmeasurable, because you have no record of what the strategy said to do, only of what you did.

How much degradation is normal

There's no universal figure — it depends on your holding period, instrument liquidity and how mechanical your rules are. But some reasonable expectations: a strategy holding positions for days on a liquid instrument with fully mechanical rules should lose relatively little. A short-holding-period discretionary strategy on a wider-spread instrument can lose a substantial proportion of its theoretical edge to slippage and interpretation alone.

The practical rule is comparative rather than absolute: **measure your own gap, then watch whether it's stable**. A consistent gap of 30% is workable — you simply size and forecast on the realised figure rather than the theoretical one. A gap that widens over time is the warning sign, because it means behavioural interference is increasing.

> WARNING — THE GAP THAT EATS THE EDGE
>
> If your theoretical expectancy is +0.15R and your live gap is 0.12R, your realised expectancy is +0.03R — technically positive, practically unprovable, and almost certainly negative after a bad run of slippage. Strategies with small theoretical edges are disproportionately destroyed by the live gap, which is another argument for pursuing fewer, larger edges rather than optimising marginal ones.

When forward results contradict the backtest

Three possibilities, in order of likelihood:

1. **The backtest was flawed.** Check the seven biases from Chapter 11 first, particularly look-ahead and cost omission. This is the most common answer and the hardest to accept.
2. **The sample is too small.** Fifty live trades cannot contradict a five-hundred-trade backtest with any statistical authority. Compute the confidence interval before concluding anything.
3. **The regime differs.** Segment both datasets by regime and compare like with like. A strategy that worked in trending conditions and is being tested in a range hasn't failed; it's out of context.

Only after eliminating all three should you conclude the strategy doesn't work. Most traders start at that conclusion and work backwards, which is why they discard so many workable systems.

The forward test log

Each row: date, setup, planned entry/stop/target/size, actual entry/exit/size, result in R theoretical (what the rules would have produced), result in R actual, adherence flag, and a one-line note on any deviation. The difference between the two R columns, summed across the sample and divided by trade count, is your execution gap in R per trade. It is one subtraction, and almost nobody performs it.

Chapter 12 — What to take with you

- [ ] Every strategy underperforms live. Expect it, measure it, don't panic at it.
- [ ] Record intent before outcome, or the gap is unmeasurable.
- [ ] Roughly 200 trades across four stages before full size.
- [ ] A stable gap is workable. A widening gap is a behavioural warning.

13

PART III — BUILDING & PROVING AN EDGE · CHAPTER THIRTEEN

Edge Decay

Why edges deteriorate, how to detect it before it costs you a year, and the response protocol that avoids both false alarms and denial.

Edges are not permanent. Some erode slowly as conditions shift; some close abruptly when the behaviour they exploited changes. The traders who survive decades are not the ones who found a permanent edge — no such thing exists — but the ones who noticed decay early enough to respond while they still had capital.

Why edges decay

MECHANISM	WHAT HAPPENS	SPEED
Crowding	Enough participants trade the same inefficiency that it stops paying	Gradual, then sudden
Regime shift	The market condition your edge depends on stops being present	Can be abrupt; often reverses later
Structural change	Market hours, tick sizes, participant composition or venue structure change	Permanent
Cost change	Spreads widen or commissions rise; a marginal edge crosses into negative	Immediate and easily missed
Adaptation	Other participants learn to anticipate the behaviour you were exploiting	Gradual

Note the distinction between **regime shift** and true decay. A trend system in a two-year range hasn't decayed — it's out of conditions, and it will likely work again. Treating a regime mismatch as decay causes traders to abandon perfectly good systems permanently, which is why Chapter 15's segmentation matters so much here.

The rolling expectancy monitor

The core detection tool. Plot expectancy over a rolling window — 50 trades is a reasonable default — alongside the confidence band derived from your full-sample parameters (Chapter 6). Ordinary variance moves within the band. Genuine decay shows as a sustained trend leaving it.

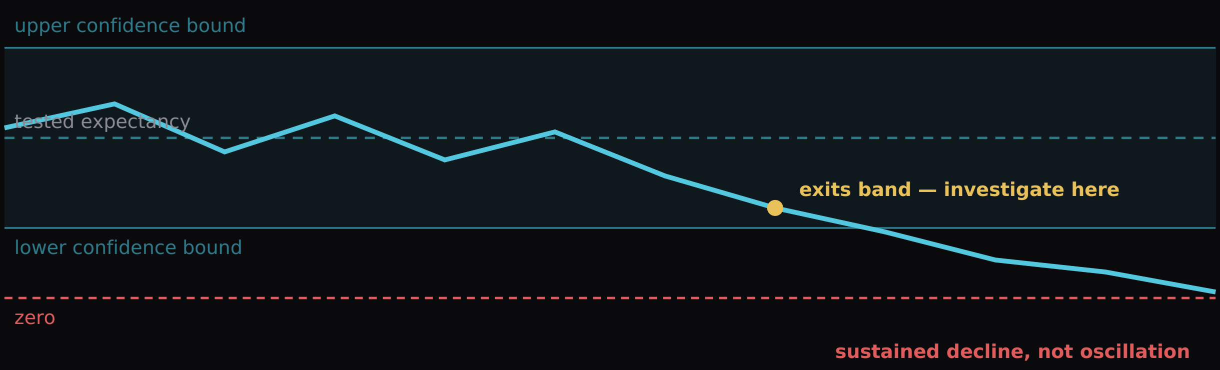


FIG 13.1 — VARIANCE OSCILLATES INSIDE THE BAND. DECAY LEAVES IT AND KEEPS GOING. THE DISTINCTION IS DIRECTION AND PERSISTENCE, NOT DEPTH.

The four monitoring metrics

Rolling expectancy alone can lag. Watch these together — several will move before expectancy does:

METRIC	WHAT A DECLINE INDICATES
Rolling expectancy (50)	The headline measure. Sustained exit below the band triggers investigation.
Win rate, rolling	Falling win rate with stable win/loss ratio points at entry quality or setup conditions changing.
Average win in R	Often the earliest warning. Targets becoming unreachable usually signals a volatility regime change before expectancy reflects it.
Setup frequency	Underrated. If your setup is appearing half as often, the conditions producing it have changed — even if the ones that do appear still work.

DRAWDOWN INSIGHT — FREQUENCY IS THE FORGOTTEN SIGNAL

A setup that still works but appears a third as often has effectively decayed as a business proposition, even though its expectancy is unchanged. Your annual return is expectancy multiplied by frequency, and traders monitor only the first term. If your opportunity count is falling, you have a problem regardless of what your expectancy is doing.

Distinguishing decay from the alternatives

Before concluding decay, eliminate the other three explanations in order:

1. **Drift.** Has adherence fallen? If yes, it's you. Nothing else can be assessed until this is resolved.
2. **Variance.** Is the decline within the simulated range for your system? Run the Chapter 7 bootstrap on your own results.
3. **Regime.** Segment by regime. Are losses concentrated in conditions your edge was never suited to?
4. **Costs.** Have spreads, commissions or financing changed? Recompute expectancy net of current costs, not the ones you assumed.

Only when all four are eliminated is decay the supported conclusion. In practice, most suspected decay resolves as one of the first three.

The response protocol

FRAMEWORK — RESPONDING TO SUSPECTED DECAY

Trigger: rolling expectancy below the lower confidence bound for 30+ consecutive trades, or two of the four monitoring metrics declining together.

1. Halve risk immediately. This costs little and buys time.
2. Run the four eliminations above and document the findings in writing.
3. Continue at half risk for a further 50 trades minimum.
4. If the cause is drift, regime or costs → fix it, restore size, no strategy change.
5. If decay is confirmed → ask whether the original reason (Chapter 9) still holds. If the reason is gone, retire the strategy. If it holds but the implementation no longer matches conditions, rebuild and re-test as a new version.
6. Never scale back up on a recovered run alone. Restore size only when rolling expectancy re-enters the band and stays there.

Building the monitor

The whole apparatus is four columns and a chart. Column one: trade number. Column two: rolling 50-trade mean of actual_R. Columns three and four: your upper and lower confidence bounds, computed once from your full-sample σ and held constant. Plot all four. That single chart replaces the entire emotional process most traders use to decide whether their system still works.

Two refinements worth adding once it's running. First, plot setup frequency on a secondary axis — as Chapter 15 showed, a falling opportunity count is a decay signal that expectancy alone will never surface. Second, mark specification version changes on the chart as vertical lines, so you can see immediately whether a deterioration began before or after you changed something. That second refinement resolves an enormous number of arguments with yourself.

WARNING — THE SUNK COST TRAP

The more time you invested developing a strategy, the harder it is to retire. Traders routinely spend years trading a decayed edge because abandoning it would mean the development was wasted. It wasn't — the process taught you how to build and test, which is transferable. The trades you take from here are the only ones that can cost you anything.

Chapter 13 — What to take with you

- [ ] No edge is permanent. Build monitoring in from the start.
- [ ] Watch four metrics — expectancy, win rate, average win, and frequency.
- [ ] Eliminate drift, variance, regime and costs before concluding decay.
- [ ] Halve risk while investigating. It is almost never wrong.

14

PART IV — PERFORMANCE ANALYTICS · CHAPTER FOURTEEN

The Metrics That Matter

What to measure, what each number actually tells you, and the popular metrics that mislead more than they inform.

Most traders track two numbers: account balance and win rate. Both are close to useless in isolation — one is contaminated by sizing and luck, the other is meaningless without the win/loss ratio. This chapter sets out the metrics professionals actually use, what each reveals, and how to read them together.

The core set

METRIC	DEFINITION	WHAT IT TELLS YOU
Expectancy (R)	Mean R per trade	Whether repetition helps or hurts. The headline number.
Standard deviation (R)	Spread of trade results	How much evidence you need and how violent the ride will be.
Expectancy ÷ σ	Edge per unit of variability	The best single measure of system quality — comparable across strategies.
Profit factor	Gross profit ÷ gross loss	Intuitive robustness check. Above 1.3 across a large sample is respectable; above 2.0 warrants suspicion of a small sample or an outlier.
Max drawdown & duration	Deepest peak-to-trough, and how long to recover	Duration is the more important half and almost nobody records it. Depth tests your capital; duration tests your conviction.
Adherence rate	% of trades following specification exactly	Whether you're measuring a strategy or measuring yourself.
Execution gap (R)	Theoretical R minus realised R, per trade	The cost of your behaviour, in the same units as your edge.
Opportunity frequency	Qualifying setups per week	Expectancy × frequency is your actual return. Both terms matter equally.

DRAWDOWN INSIGHT — THE SINGLE BEST QUALITY MEASURE

Expectancy divided by standard deviation is the number to optimise. A system with +0.2R expectancy and 1.0 σ (ratio 0.20) is materially better than one with +0.25R and 1.8 σ (ratio 0.14), despite the second having higher headline expectancy — because the first is smoother, provable in fewer trades, and can be sized more aggressively for the same drawdown. Most traders compare only the first number and reach the wrong conclusion.

The vanity metrics

METRIC	WHY IT MISLEADS
Win rate alone	Meaningless without the win/loss ratio. A 70% win rate with 0.5R wins and 1.5R losses is a losing system, and it feels wonderful while it drains you.
Account balance	Contaminated by deposits, withdrawals, sizing changes and luck. It's an outcome, not a measurement.
Total pips or points	Ignores position size entirely. You can gain pips and lose money without contradiction.
Best month / best trade	Selects for outliers. Tells you what luck looks like, not what your system does.
Consecutive winning days	Sequence artefacts. Feels significant, contains no information.
Return without drawdown	Any return figure quoted without the drawdown that produced it is incomplete to the point of being misleading. 40% with a 45% drawdown is a worse system than 15% with a 6% one.

Drawdown duration — the metric nobody tracks

Depth gets all the attention; duration does the damage. A 15% drawdown recovered in three weeks is a minor event. The same 15% taking eight months to recover is what makes traders abandon working systems, change specifications mid-sample, and increase risk to accelerate recovery. Record both, and know your system's expected duration from simulation — because eight months feels like failure unless you knew in advance it was possible.

Reading metrics together

Individual numbers mislead; combinations diagnose. Some readings worth recognising:

PATTERN	DIAGNOSIS
High expectancy, high σ, long drawdowns	Low win rate / high payoff system. Legitimate, but needs a large sample and strong conviction. Size conservatively.
Positive R, negative currency	Sizing failure (plateau 4). Losses concentrated in oversized trades.
Good adherence, declining expectancy	Genuine decay or regime mismatch. Run Chapter 13's eliminations.
High theoretical, low realised	Execution gap (plateau 3). The strategy needs nothing.
High profit factor, few trades	Almost certainly outlier-driven. Run the outlier removal test.
Stable expectancy, falling frequency	Business-level decay. The edge works but no longer pays enough to matter.

The reporting cadence

FRAMEWORK — WHAT TO REVIEW, AND WHEN

Daily (2 min): trades logged, adherence flags set. Data integrity only — no analysis, no conclusions.
Weekly (20 min): R total, adherence rate, any rule breaches read individually, one written intention for next week.
Monthly (60 min): rolling expectancy vs confidence band, the four decay metrics, execution gap, segmentation by setup and session.
Quarterly (half day): full audit page (Chapter 3), simulation refresh with updated parameters, risk budget review, plateau re-diagnosis, and any specification version changes.

Note where the decisions live: nothing is changed daily or weekly. Structural change happens monthly at the earliest and quarterly by default. This is deliberate — decisions made at high frequency are made on noise.

WARNING — METRICS ARE NOT A SUBSTITUTE FOR SAMPLE SIZE

Every metric in this chapter requires an adequate sample to mean anything. A profit factor of 2.4 across 20 trades is not a good system; it is a number. Compute your confidence interval first, then read your metrics. Sophisticated analysis applied to insufficient data produces confident nonsense, and confident nonsense is more dangerous than admitted ignorance.

Chapter 14 — What to take with you

- [ ] Optimise expectancy ÷ standard deviation, not expectancy alone.
- [ ] Record drawdown duration, not just depth. Duration is what breaks people.
- [ ] Never quote a return without the drawdown that produced it.
- [ ] Structural decisions monthly or quarterly. Daily and weekly are for data, not conclusions.

15

PART IV — PERFORMANCE ANALYTICS · CHAPTER FIFTEEN

Segmentation

Cutting your results by setup, session, day, regime and holding time — the analysis that most often turns a flat system into a profitable one.

Aggregate results conceal almost everything interesting. A trader with flat overall performance very often has one strongly profitable segment and two loss-making ones, and has been trading all three with equal enthusiasm for years. Segmentation is how you find out — and it is, in practice, the single highest-yield analysis in this book.

The axes worth cutting

AXIS	QUESTION IT ANSWERS	TYPICAL FINDING
Setup	Which of my specifications actually pays?	One setup carries the account; another has been quietly negative for a year.
Session / time	When am I actually good?	Performance concentrated in one session; another session is a consistent leak.
Day of week	Is there a structural or behavioural pattern?	Fridays worse (position-closing behaviour); Mondays worse (weekend-gap positioning).
Market regime	What conditions does my edge require?	Strong in trend, negative in range — an enormously common and highly actionable finding.
Volatility state	Does my edge need movement, or calm?	Targets unreachable in low volatility; stops too tight in high volatility.
Holding time	Am I exiting too early or too late?	Winners cut short — the most common single finding in retail data.
Position in sequence	Does my performance change after a loss?	Trades taken immediately after a loss are significantly worse. Behavioural, and fixable with a rule.

DRAWDOWN INSIGHT — THE POST-LOSS SEGMENT

Cut your results by "trades taken within an hour of a losing trade" versus everything else. For a large proportion of traders this single cut reveals a substantial negative expectancy in the first group — meaning a mandatory cooling-off rule would improve overall results more than any strategy change. It costs nothing, requires no market knowledge, and takes ten minutes to test on your existing data.

A worked segmentation

An illustrative dataset of 240 trades with an unremarkable aggregate expectancy of +0.04R — the kind of result that leads traders to conclude their system doesn't work:

SEGMENT	TRADES	EXPECTANCY	READ
Setup A · trending regime	78	+0.31R	The actual edge. Everything else is diluting it.
Setup A · ranging regime	64	-0.18R	Same setup, wrong conditions. A filter fixes this.
Setup B · all regimes	52	+0.02R	Indistinguishable from zero. Consuming attention and costs.
Post-loss trades	46	-0.22R	Behavioural. Fixed with a cooling-off rule, not a strategy change.

The aggregate said "marginal system, probably abandon." The segments say something completely different: there is a genuinely good edge here, being masked by a regime mismatch, a redundant setup and a behavioural leak. Adding a regime filter, retiring Setup B and imposing a post-loss pause would transform this trader's results without touching the thing that actually works.

The sample size problem in segmentation

Here is the catch, and it is a serious one. Every time you cut the data, each resulting segment is smaller and therefore noisier. Split 240 trades four ways and you have segments of 50–80 trades — well below the threshold needed to establish anything with confidence (Chapter 6). Worse, if you test enough segmentations, some will look significant purely by chance. This is the multiple-comparisons problem again, and segmentation is where it does the most damage.

> WARNING — HOW TO SEGMENT WITHOUT FOOLING YOURSELF
>
> **Form the hypothesis first.** "I expect this trend setup to underperform in ranges" is a prediction you can test. Cutting the data twenty ways and reporting the interesting slices is not analysis, it's data mining, and it will reliably produce a finding that evaporates the moment you trade it. Decide what you're testing before you look, limit yourself to a handful of pre-specified cuts, and treat any surprise discovery as a hypothesis for future data rather than a conclusion.

Acting on segmentation

> FRAMEWORK — FROM FINDING TO CHANGE
>
> **1. Is the segment large enough?** Compute the confidence interval for that segment specifically. If it crosses zero, you have a hypothesis, not a finding.
> **2. Is there a mechanism?** Can you explain why this segment differs, in terms of the four sources from Chapter 9? A finding without a mechanism is probably noise.
> **3. Was it predicted?** Hypothesis-first findings are far more trustworthy than discovered ones.
> **4. Test it forward.** Apply the proposed filter prospectively for 50+ trades before making it permanent.
> **5. Version the specification** when you adopt it, and start a fresh sample (Chapter 10).

The exit analysis

One specific segmentation deserves separate mention because it so reliably produces value. For every trade, record the maximum favourable excursion — how far the trade went in your favour before you exited. Then compare it with your actual exits.

The typical retail finding: winners are exited at a fraction of their available move while losers are held to full stop. That asymmetry is not a strategy problem — it's loss aversion expressing itself through your exit decisions, and it can be worth a substantial part of your edge. The fix is mechanical exits rather than better judgement, because judgement is precisely what's failing.

The segmentation you should run first

If you only run one, run **setup × regime**. It's the cut most likely to contain a large, mechanistically explicable effect, and it produces the most actionable output — a filter. Second priority is post-loss versus normal, because it's behavioural, quick to test, and fixable with a rule rather than a redesign.

Chapter 15 — What to take with you

- [ ] Aggregate results hide everything. Flat systems usually contain a good one.
- [ ] Form the hypothesis before you cut. Discovered findings are usually noise.
- [ ] Each segment needs its own confidence interval. Small segments prove nothing.
- [ ] Run setup × regime first, then post-loss. Both are high-yield and quick.

16

PART IV — PERFORMANCE ANALYTICS · CHAPTER SIXTEEN

The Execution Gap

Measuring the distance between the strategy you designed and the one you actually traded — and recovering the edge sitting inside it.

You have two strategies: the one written in your specification and the one visible in your trade history. The difference between them is the execution gap, it is measurable in R, and for most struggling traders it is larger than any edge they could realistically find by researching new setups.

The measurement

For every trade, record two results: **theoretical R** — what the specification would have produced with perfect execution — and **actual R**. The gap is the difference, averaged across the sample. It requires no statistics beyond subtraction, and its value comes from decomposing it.

Illustrative decomposition: a good strategy reduced to a marginal one, entirely by execution.

FIG 16.1 — THE EXECUTION WATERFALL. EACH COMPONENT IS SEPARATELY MEASURABLE AND SEPARATELY FIXABLE. RECOVERING HALF OF THIS GAP IS WORTH MORE THAN MOST STRATEGY RESEARCH.

The five components

COMPONENT	HOW TO MEASURE IT	HOW TO REDUCE IT
Slippage	Planned entry/exit price vs actual fill, converted to R	Limit orders where appropriate; avoid the most volatile moments; check whether one instrument or session dominates the cost
Missed trades	Valid signals not taken, evaluated at their theoretical outcome	Alerts, pending orders, defined session presence. Log every miss.
Early exits	Actual exit vs specification exit on winning trades	Mechanical exits; remove the decision from the moment
Widened stops	Actual loss vs 1R on losing trades	Hard stops placed at entry; never adjustable except in your favour
Unauthorised trades	Trades with no matching specification, evaluated separately	Pre-trade checklist; hard daily limits

Reading the decomposition

Each component points at a different fix, which is why decomposition matters more than the headline number:

- **Slippage dominant** → structural. Wrong instrument, session, order types, or too short a holding period relative to costs. Not a discipline issue.
- **Missed trades dominant** → availability. Your strategy needs presence you don't have. Automate entries, use pending orders, or move to a compatible timeframe.
- **Early exits dominant** → loss aversion. The most common single component, and the most responsive to mechanisation.
- **Widened stops dominant** → the most dangerous pattern. This is the behaviour that converts controlled losses into account-ending ones. Treat it as a hard-stop-only problem immediately.
- **Unauthorised trades dominant** → you are not currently running a strategy. Return to plateau 1 or 3 and resolve that before any further analysis.

DRAWDOWN INSIGHT — WHERE THE MONEY ACTUALLY IS

Consider the arithmetic. Finding a new strategy with +0.1R expectancy takes months of research, a few hundred trades to validate, and may fail entirely. Closing half of a 0.2R execution gap delivers the same +0.1R, is available immediately, requires zero new market knowledge, and has a near-certain probability of working. Traders spend overwhelmingly more time on the first. This is arguably the largest single misallocation of effort in retail trading.

Tracking it over time

Plot execution gap monthly alongside expectancy. Three patterns are worth recognising:

PATTERN	INTERPRETATION
Gap stable, expectancy stable	Healthy. Forecast and size using realised figures, not theoretical ones.
Gap widening	Behavioural deterioration, usually stress-related and usually preceding a drawdown rather than following it. This is an early warning system.
Gap narrowing	Genuine skill development — and the most reliable evidence of progress available to you, considerably more so than P&L.

The mechanisation principle

Every component of the gap reduces when the decision is removed rather than improved. Hard stops placed at entry eliminate widened stops entirely. Pre-set take-profits eliminate early exits. Pending orders eliminate missed trades. A pre-trade checklist eliminates unauthorised trades.

The pattern is worth stating explicitly: **you do not close the execution gap by trying harder, but by having fewer decisions available to make.** Discipline is a design property, not a character trait.

FRAMEWORK — THE MONTHLY GAP REVIEW

1. Compute total theoretical R and total realised R for the month. **2.** Divide the difference by trade count to get gap per trade. **3.** Attribute the gap across the five components. **4.** Identify the largest single component. **5.** Implement one mechanical control targeting it. **6.** Measure next month. One component at a time — attempting all five simultaneously reliably fixes none.

A worked case

A trader runs a specification with a tested expectancy of +0.26R and cannot understand why their account is flat across 180 trades. Their realised expectancy is +0.03R, so the gap is 0.23R per trade. Decomposed: slippage 0.03R, missed trades 0.02R, early exits 0.11R, widened stops 0.06R, unauthorised trades 0.01R.

Early exits alone account for nearly half the gap. The trader's instinct — and the industry's default advice — would be to find a better strategy. The data says something completely different: install a mechanical take-profit, remove the exit decision entirely, and recover roughly 0.11R per trade. Across 180 trades at 1% risk that is a materially different year, achieved without learning anything new about markets.

Note also what the decomposition rules out. Slippage is small, so the instrument and session are appropriate. Missed trades are small, so availability isn't the constraint. Unauthorised trades are negligible, so discipline in the broad sense is intact. The problem is narrow, specific and mechanically fixable — which is the whole reason to decompose rather than simply noting that a gap exists.

Chapter 16 — What to take with you

- [ ] The gap is measurable with subtraction, and it is usually larger than you'd guess.
- [ ] Decompose it — each component has a different, specific fix.
- [ ] Closing the gap is cheaper and more certain than finding a new edge.
- [ ] Remove decisions rather than improving them. Discipline is design.

17

PART IV — PERFORMANCE ANALYTICS · CHAPTER SEVENTEEN

Building Your Analytics System

The practical build — data structure, the calculations, and the dashboard that turns a trade log into a decision-making instrument.

Everything in this book depends on having data in a usable structure. This chapter is the practical build: what the table looks like, what to compute from it, and how to present the output so it drives decisions rather than sitting in a file you never open.

The data structure

One row per trade, one sheet. Resist the urge to split across tabs — analysis becomes far harder and the friction means you'll stop doing it.

GROUP	FIELDS
Identity	trade_id · date · time · instrument · spec_name · spec_version
Context	session · day_of_week · regime · volatility_state · preceded_by_loss (Y/N)
Intent	planned_entry · planned_stop · planned_target · planned_size · planned_risk_currency
Reality	actual_entry · actual_exit · actual_size · exit_reason · max_favourable_excursion
Results	theoretical_R · actual_R · currency_result · costs
Quality	adherence (Y/N) · deviation_note · screenshot_link

Around 25 fields. Roughly two minutes per trade to complete honestly. If that feels like too much, you are trading too frequently for the amount of analysis you can support — which is itself a finding worth acting on.

> WARNING — RECORD INTENT BEFORE OUTCOME
>
> The intent fields must be filled at decision time, not reconstructed afterwards. Once you know the result, you cannot honestly recall what you planned — you'll record what would have been sensible. Everything in Chapters 12 and 16 depends on this discipline, and no amount of care after the fact substitutes for it.

The calculation layer

From those columns, compute:

- **Expectancy** — mean of actual_R, filtered to adherent trades, per spec_version
- **Standard deviation** of actual_R, and the derived confidence interval $E \pm 1.96\sigma/\sqrt{n}$
- **Rolling expectancy** over the last 50 trades, plotted against the confidence bounds
- **Execution gap** — mean of (theoretical_R − actual_R), and the same split by exit_reason to get the component decomposition
- **Adherence rate**, overall and rolling
- **Risk consistency** — standard deviation of planned_risk_currency ÷ its mean
- **Segment expectancies** — pivot by regime, session, day, preceded_by_loss
- **Drawdown depth and duration** from the running equity series
- **Frequency** — qualifying setups per week, including the ones you didn't take

The bootstrap sheet

A separate sheet that resamples your actual_R column with replacement to produce the distribution work from Chapter 7. Build it once and it serves for the life of your trading: sample n results at random, compound at your risk level, record final equity and maximum drawdown, repeat several thousand times, then read the 5th, 50th and 95th percentiles of each. Refresh quarterly as your sample grows.

The dashboard

One screen, four blocks. It should answer four questions without any further clicking:

BLOCK	ANSWERS	CONTENTS
1 · Am I executing?	Is my behaviour matching my plan?	Adherence rate (rolling 50), execution gap per trade, risk consistency ratio
2 · Is the edge healthy?	Variance, decay, or fine?	Rolling expectancy vs confidence band, win rate, average win in R, setup frequency
3 · Where does it come from?	Which segments pay?	Expectancy by setup × regime, by session, and post-loss vs normal
4 · What am I risking?	Am I sized correctly?	Current drawdown vs simulated distribution, risk per trade vs calculated ceiling, sample size vs requirement

DRAWDOWN INSIGHT — BUILD IN ORDER

Do not attempt to construct all of this at once; you'll abandon it. Build block 1 first and use it for a month — adherence and execution gap alone will change your trading more than the rest combined. Add block 2 when you have 100 trades. Add block 3 at 250. Block 4 last. An analytics system you actually maintain beats a comprehensive one you built and stopped updating.

Tooling

A spreadsheet handles everything here, including the simulation. Specialist software often makes things worse by hiding the calculation — you learn more building the bootstrap yourself than clicking a button that produces one.

Data integrity

FRAMEWORK — KEEPING THE DATA TRUSTWORTHY

1. Same day, always. Log trades the day they close. Backfilling produces reconstructed intent, which is worthless.
2. Reconcile monthly against broker statements. Missing trades are almost never random — they cluster around your worst behaviour.
3. Never edit history. If a row is wrong, add a correction row with a note. An editable record becomes a flattering one.
4. Flag reconstructed rows and exclude them from adherence and gap analysis entirely.
5. Back it up. Years of honest records are genuinely irreplaceable, and they are the most valuable asset you will build as a trader.

Chapter 17 — What to take with you

- ☐ One row per trade, ~25 fields, intent recorded before outcome.
- ☐ Build block 1 first. Adherence and gap alone will change your trading.
- ☐ A spreadsheet does everything here, including simulation.
- ☐ Never edit history. Add corrections instead.

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PART V — SCALING CONSISTENCY · CHAPTER EIGHTEEN

From One Setup to a Portfolio

Adding edges without adding correlated risk — when to expand, how to test additions, and why most traders diversify into the same trade repeatedly.

A single setup on a single instrument has a hard ceiling: its frequency. Expanding is how you raise that ceiling — but expansion done carelessly reduces returns and increases risk simultaneously, which is a genuinely impressive achievement and a very common one.

When you are ready to expand

- ☐ Your existing specification has 200+ trades and expectancy confidently above zero.
- ☐ Adherence has held above 95% for at least three months.
- ☐ Your execution gap is measured and stable.
- ☐ You have surplus attention — the existing system genuinely doesn't consume all of it.
- ☐ The constraint you're solving is frequency, not performance. This is the important one.

That last point deserves emphasis. Adding a second setup because the first isn't working is not diversification — it's abandonment with extra steps, and it resets your sample to zero on both. Expand only from a position of strength.

The correlation problem

Three setups that all profit from trend continuation are one setup with three entry techniques. When trends stop, all three stop together, and your drawdown is three times what you modelled for one. Meanwhile you've tripled your monitoring load and divided your sample three ways.

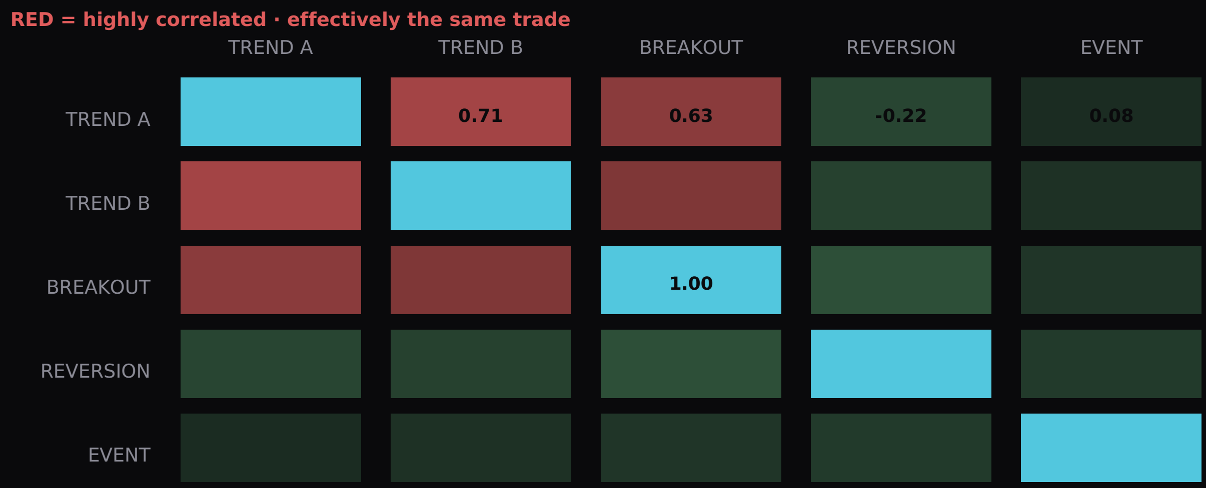

FIG 18.1 — ILLUSTRATIVE CORRELATION MATRIX. TREND A, TREND B AND BREAKOUT ARE ONE EXPOSURE WEARING THREE NAMES. REVERSION AND EVENT GENUINELY DIVERSIFY.

Testing correlation between your own setups

You don't need sophisticated statistics. Group your trades into calendar weeks, compute total R per setup per week, and compare the series. If two setups have good weeks and bad weeks together, they're correlated regardless of how different they look on a chart. A more direct question: during my worst drawdown, what did each setup do? If they all lost together, you have one exposure.

What genuine diversification looks like

AXIS	DIVERSIFIES WELL	DOESN'T
Direction of edge	Trend continuation vs mean reversion — they profit from opposite conditions	Three trend entries with different triggers
Timeframe	Intraday vs multi-day — different drivers, different noise	15-minute and 30-minute versions of the same idea
Instrument class	FX vs equity index vs commodity — genuinely different drivers	Three correlated FX majors in a risk-on/risk-off market
Edge source	Behavioural vs structural (Chapter 9)	Two behavioural edges exploiting the same bias

> WARNING — CORRELATION RISES EXACTLY WHEN IT HURTS
>
> Instruments that look independent in normal conditions frequently move together during stress. Correlations converge in crises, which means your diversification is at its weakest precisely when you most need it. Assume your worst-case combined drawdown is meaningfully worse than the sum of independent simulations, and size accordingly.

The addition protocol

> FRAMEWORK — ADDING A SETUP WITHOUT DAMAGING WHAT WORKS
>
> **1. Specify it fully** before trading it (Chapter 10), including the edge rationale from Chapter 9.
> **2. Test correlation** against your existing setups using historical or backtested weekly R. If correlation is high, either reject it or accept it as a variant of the existing setup rather than an addition.
> **3. Backtest and forward-test** it independently to the standard of Chapters 11 and 12. It must earn its place on its own evidence.
> **4. Run it at minimum size** alongside the existing system for 50 trades. Critically: **monitor the existing system's adherence** during this period. If it deteriorates, the addition is costing you more than it's contributing, and the correct move is to withdraw it.
> **5. Allocate risk deliberately** (Chapter 19) rather than trading both at full size.
> **6. Review at 100 trades** and decide: keep, retire, or continue testing.

The attention constraint

The binding limit for most traders isn't capital or ideas — it's attention. Two well-executed setups comfortably outperform five poorly-monitored ones, and the failure mode is subtle: expectancy doesn't collapse, adherence does.

The test is simple: if adding a setup reduces adherence on the existing one, the addition has negative expected value regardless of its standalone performance — it degrades a proven edge to fund an unproven one.

The realistic end state

It is worth being clear about what a mature retail portfolio actually looks like, because the imagined version is usually far more elaborate than the effective one. Two to four specifications. One or two instruments each. Genuine diversification across perhaps two axes rather than five. Total open risk capped well below what any single specification would justify on its own.

That is a smaller operation than most traders expect to end up with, and it outperforms the sprawling alternative for a straightforward reason: every component is individually proven, individually monitored, and individually retirable. When one decays you can identify it, halve its allocation, and continue — because the others are separately measured. A trader running eleven loosely-defined setups cannot do any of that, and when their results deteriorate they have no way to locate the cause.

Expansion, done properly, is therefore less about adding and more about qualifying. Most candidate setups should fail the addition protocol. If everything you test passes, your testing standard is the problem.

Chapter 18 — What to take with you

- [ ] Expand to solve frequency, never to escape poor performance.
- [ ] Correlated setups are one exposure. Test with weekly R, not with your eyes.
- [ ] Correlations converge under stress. Size for the worst case, not the average.
- [ ] If a new setup damages adherence on the old one, it has negative value.

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PART V — SCALING CONSISTENCY · CHAPTER NINETEEN

Risk Budgeting

Allocating risk across setups by evidence rather than enthusiasm — and the disciplined path to trading larger.

Once you have more than one setup, a new question appears: how much risk should each receive? Most traders answer it by feel, which reliably means the newest and most exciting setup gets disproportionate capital while the proven one gets neglected. Risk budgeting replaces that with arithmetic.

The portfolio risk budget

Start from the top. Decide your total acceptable portfolio risk — the maximum you're prepared to have exposed at any moment, and the maximum drawdown you'll accept before intervening. Then allocate that budget across setups, rather than sizing each one independently and discovering your aggregate exposure by accident.

BUDGET LEVEL	TYPICAL STRUCTURE
Portfolio	Maximum total open risk at any moment (e.g. 3% of account across all positions); maximum portfolio drawdown before mandatory review
Setup	Each setup's share of the total, based on evidence quality and edge size
Trade	Per-trade risk within each setup's allocation, from the Chapter 8 constraints
Correlation overlay	A cap on combined exposure to correlated setups, treating them as one position for limit purposes

Allocating by evidence

A defensible allocation weights three things: **edge quality** (expectancy ÷ standard deviation, from Chapter 14), **evidence strength** (sample size relative to the Chapter 6 requirement), and **diversification contribution** (how uncorrelated it is with everything else).

SETUP	E ÷ Σ	SAMPLE VS REQUIREMENT	CORRELATION WITH CORE	ALLOCATION
Core trend	0.20	420 / 207 ✓	–	55%
Reversion	0.16	260 / 190 ✓	-0.22	30%
Event	0.22	70 / 180 ✗	0.08	15%

Illustrative. Note the Event setup: the best edge quality of the three, and the smallest allocation — because the evidence isn't there yet. Allocation follows proof, not promise.

> DRAWDOWN INSIGHT — THE DISCIPLINE THIS ENFORCES
>
> Weighting by evidence rather than by attractiveness is what stops a trader moving capital to whatever performed best last month. Recent performance is the single worst allocation signal available — it's dominated by variance, and chasing it systematically means buying every setup at its local peak.

Scaling up

Increasing size is the point at which many otherwise-competent traders come undone, because the behavioural constraint from Chapter 8 binds long before the mathematical one. A position that is 3% of your account produces identical emotions regardless of whether that's £30 or £3,000 — right up until it doesn't.

> FRAMEWORK — THE SCALING LADDER
>
> **Increase size only when all four conditions hold:**
>
> **1.** Sample exceeds the Chapter 6 requirement at the current size.
> **2.** Adherence above 95% for three consecutive months.
> **3.** Execution gap stable or narrowing.
> **4.** The proposed size passes all four Chapter 8 constraints, including the behavioural one.
>
> **Increase in steps of no more than 25–50%**, then trade 50 trades at the new size before considering the next step. If adherence falls or the execution gap widens at the new size, step back down immediately — that is a behavioural limit revealing itself, and it is information, not failure.

Scaling down

Equally important and far less practised. Reduce risk when: your rolling expectancy leaves its confidence band; adherence falls below 90%; the execution gap widens materially; you're in unfamiliar market conditions; or your personal circumstances have changed in ways that affect concentration.

Reducing size is not an admission of failure — it's the cheapest available insurance. Halving risk during uncertainty costs you half the upside on a system you're currently unsure about, which is a rational trade. The traders who survive decades are conspicuously willing to trade small.

The compounding decision

Whether to size from a growing account balance or a fixed base is a genuine choice with consequences. Compounding accelerates growth and also accelerates drawdowns, because losses arrive at the larger size. A practical compromise many professionals use: size from a base that's updated periodically — quarterly, say — rather than continuously. You capture most of the compounding benefit without your position size tracking every equity fluctuation, and it removes a source of daily decision-making.

Withdrawals as risk management

Taking money out reduces the capital at risk, and there's an argument for scheduled withdrawals as a discipline mechanism: it converts paper equity into realised results and prevents the account from growing to a size where a normal drawdown becomes psychologically intolerable. It also forces the question of whether trading is producing anything real. Whatever you decide, decide it in advance and schedule it — withdrawal decisions made after a good month are made for the wrong reasons.

> ### Chapter 19 — What to take with you
>
> - [ ] Budget risk from the portfolio down, not the trade up.
> - [ ] Allocate by edge quality, evidence and diversification — never by recent performance.
> - [ ] Scale up in steps, with 50 trades at each. Step back the moment adherence slips.
> - [ ] Willingness to trade small is a survival characteristic.

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PART V — SCALING CONSISTENCY · CHAPTER TWENTY

The Operating Rhythm

The daily, weekly, monthly and quarterly cycle that turns everything in this book into a system you actually run.

Knowledge doesn't produce consistency; process does. This chapter assembles the entire book into an operating rhythm — what you do daily, weekly, monthly and quarterly, and crucially, at which level each type of decision is permitted to be made.

The decision hierarchy

The most important principle here is that **decision authority is tied to cadence**. Fast cycles handle execution; slow cycles handle structure. Making structural decisions on a daily cadence is how traders end up with a different system every fortnight.

DAILY	WEEKLY	MONTHLY	QUARTERLY
15 MIN	20 MIN	60 MIN	HALF DAY
· Calendar check	· R total	· Rolling expectancy	· Full audit page
· Mark levels	· Adherence rate	· Four decay metrics	· Simulation refresh
· Name the regime	· Read every breach	· Execution gap	· Risk budget review
· Execute spec	· One intention	· Segmentation	· Plateau re-diagnosis
· Log trades	· Prep next week	· One control added	· Spec versioning
· Set adherence flags			· Scale up / down
NO decisions permitted	Behavioural fixes only	Filters & controls	Structural changes

FIG 20.1 — THE OPERATING RHYTHM. NOTE THE BOTTOM LINE OF EACH COLUMN: AUTHORITY INCREASES AS CADENCE SLOWS. NOTHING STRUCTURAL IS EVER DECIDED IN A TRADING SESSION.

> DRAWDOWN INSIGHT — WHY THE HIERARCHY IS THE MECHANISM
>
> Almost every destructive trading decision is a structural decision made at daily cadence: changing the strategy mid-drawdown, increasing size after a good run, adding an instrument out of boredom. The rhythm doesn't make you more disciplined — it makes those decisions unavailable at the moment you're most likely to make them badly. That is the same principle as a hard stop, applied to strategy rather than to price.

The daily cycle in detail

Pre-session (10 min): economic calendar, mark levels, name the regime, note any exclusions in force today, confirm risk per trade. **Session:** execute the specification. Nothing else. **Post-session (5 min):** log every trade including intent fields, set adherence flags honestly, note anything for the weekly review. No analysis, no conclusions, no adjustments — the daily cycle produces data, not decisions.

The weekly cycle in detail

Twenty minutes, same time each week. Total the week in R. Compute adherence rate. **Read every non-adherent trade individually** — not the aggregate, the actual trades, one at a time, and note what was happening. Identify the single behaviour that cost most. Write one specific intention for next week. Then prepare next week's levels and calendar.

One intention, not five. The traders who improve fastest change one thing at a time and let it embed before adding another.

The monthly cycle in detail

An hour. Update rolling expectancy against the confidence band. Check the four decay metrics from Chapter 13. Compute and decompose the execution gap. Run your pre-specified segmentations. Then implement **one** mechanical control targeting the largest identified problem — and only one, so that next month's data can tell you whether it worked.

The quarterly cycle in detail

Half a day, and treat it as a genuine review rather than a formality. Produce the one-page audit from Chapter 3. Refresh your bootstrap simulation with the updated sample. Review the risk budget and allocations. Re-run the plateau diagnosis from Chapter 2 — plateaus change, and traders frequently solve one without noticing they've moved to the next. Make any specification version changes here, with written rationale. Decide on scaling up or down. File the audit page alongside the previous ones.

> FRAMEWORK — THE ANNUAL REVIEW
>
> Once a year, additionally: compare the four quarterly audit pages side by side and look at the trajectory rather than any single quarter. Has adherence trended up? Has the execution gap narrowed? Has expectancy stabilised? Has your sample grown enough to support the confidence you're operating with?
>
> Then compute the honest business numbers: total costs as a percentage of capital, hours invested, realised versus paper returns, and what those hours produced per hour. This is the review that answers whether trading is working as an activity, not merely whether last quarter was green.

Protecting the rhythm

The reviews are the first thing to disappear under stress — precisely when they're most needed, because a trader in drawdown is a trader most likely to make an unforced structural error. Schedule them as fixed commitments. A missed monthly review is a data point about your state, not a scheduling accident, and it's worth recording as such.

Chapter 20 — What to take with you

- ☐ Decision authority scales with cadence. No structural changes in a session.
- ☐ Daily produces data. Weekly fixes behaviour. Monthly adds controls. Quarterly changes structure.
- ☐ One intention weekly, one control monthly. Change one thing at a time.
- ☐ Missed reviews are a symptom. Record them.

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PART V — SCALING CONSISTENCY · CHAPTER TWENTY-ONE

Stop, Pivot, or Push

Making the biggest decision honestly — from evidence rather than from hope, exhaustion or sunk cost.

At some point every trader faces a genuine decision about whether to continue. It is usually made badly — in the middle of a drawdown, on emotion, at the worst possible moment for clear thinking. This chapter is about making it properly, and about deciding in advance what evidence would change your mind.

The three honest outcomes

DECISION	WHEN IT'S CORRECT	WHAT IT LOOKS LIKE
Push	Adherence high, expectancy positive and stable, execution gap narrowing, sample growing. The system works and you're running it.	Continue, scale according to Chapter 19, expand only for frequency.
Pivot	Adherence high, execution good, but the edge is genuinely absent or decayed with the underlying reason gone.	Return to Chapter 9, find a different reason, specify, test, and rebuild. Keep the process, replace the strategy.
Stop	Multiple honest attempts, adherence persistently below what the activity requires, or the costs — financial, emotional, temporal — exceed any plausible benefit.	Stop trading. Keep the analytical skills, which transfer.

DRAWDOWN INSIGHT — DECIDE THE CRITERIA WHILE CALM

Write down, now, what evidence would make you stop. A total loss figure. A time horizon. An adherence threshold you can't sustain. A point at which the opportunity cost is no longer defensible. Written while calm, these are rational. Constructed during a drawdown, they will be either far too lenient (because you're hoping) or far too harsh (because you're demoralised). This is exactly the same principle as placing a stop loss before entering a trade.

The evidence for each

Run this against your quarterly audit pages rather than against how the last fortnight felt:

QUESTION	PUSH	PIVOT	STOP
Adherence across 200+ trades	>95%	>95%	<85% persistently
Expectancy, adherent trades	Positive, CI excludes zero	Zero or negative	Never established
Execution gap trend	Stable or narrowing	Stable	Widening
Edge rationale (Ch 9)	Holds	No longer holds	Never articulated
Trajectory across quarters	Improving	Flat with good process	Flat with poor process

Note that the Stop column is characterised by process failure, not by losses. Losing money while running a sound process is a pivot situation. Losing money while unable to run any process at all is a stop situation, and the distinction matters enormously — one is a strategy problem, the other is a fit problem.

Stopping is a legitimate outcome

There is no shame in it. Trading requires an unusual combination of analytical discipline, emotional regulation and patience with ambiguity, sustained over years, for uncertain reward. Concluding it isn't for you — particularly concluding it early, on evidence, before it has cost you significantly — is an intelligent decision made well.

It should also be said plainly: stop, at minimum temporarily, if you are trading money you cannot afford to lose, if it is affecting your sleep, work or relationships, if you are concealing losses from people close to you, or if you find you cannot stop when you planned to. Those are not performance signals, they're welfare signals, and they override every analytical consideration in this book. Free confidential support is available in the UK from GamCare.

What you keep either way

The analytical apparatus you've built transfers almost entirely. Specifying a hypothesis precisely. Distinguishing signal from noise. Understanding sample size, variance, and why your intuitions about randomness are wrong. Measuring the gap between intention and behaviour. Making decisions from evidence rather than narrative. Knowing that a good decision and a good outcome are different things.

Those skills are worth having in any business, any investment decision, any domain with uncertainty in it — which is all of them. The traders who stop after building this apparatus properly have not wasted their time; they've acquired a genuinely useful way of thinking and paid a tuition fee for it.

If you push

Then the work from here is unglamorous and continuous: execute the specification, log honestly, run the rhythm, close the execution gap, monitor for decay, expand only for frequency, scale only on evidence. There's no further destination — professional trading is this, indefinitely, with the numbers getting larger and the process staying identical.

That's the actual answer to consistency, and it's why so few reach it. Not because the framework is difficult to understand — you've just read it — but because running it requires doing unremarkable things carefully, for years, with no applause and frequent doubt. The measurement capability is what makes that bearable: when you can tell variance from decay from drift, drawdowns stop being existential and become information. That's the whole difference.

Chapter 21 — What to take with you

- [ ] Write your stop criteria while calm, exactly as you'd place a stop loss.
- [ ] Losses with good process = pivot. No process at all = stop.
- [ ] Stopping is a legitimate, intelligent decision. Welfare signals override everything.
- [ ] The analytical apparatus transfers. It was never wasted.

CASE

APPENDIX — WORKED EXAMPLE

A Complete Diagnosis

The entire framework applied end to end to one trader's data — from "nothing works any more" to a specific, evidenced decision.

This appendix runs the whole book against a single illustrative dataset. The trader, the numbers and the outcome are constructed to demonstrate method — but the shape of the problem, and the sequence of surprises, is the one this analysis produces most often in practice.

The presenting problem

A trader with three years' experience. Two profitable quarters last year, four flat-to-negative quarters since. They describe the problem as "my edge stopped working" and have spent the last two months researching replacement strategies. They trade a pullback continuation setup on one index and one FX major, plus what they describe as "a breakout thing I take sometimes."

Their proposed solution: adopt a new strategy. Their actual diagnosis, as we'll see, is nothing to do with the strategy at all.

Step 1 — Build the dataset (Chapter 3)

Broker statements yield 312 trades over 14 months. Entries, exits, sizes and dates are recoverable. Setup name, adherence and regime are not — the journal covers only the first 90 trades, and there is a conspicuous four-week gap in it that corresponds exactly to their worst drawdown.

The 90 journaled trades become the primary dataset. The remaining 222 are flagged as reconstructed and used only for cost, sizing and drawdown analysis, never for adherence or gap work. **This distinction is not pedantry.** Mixing them would have produced an adherence figure that was flattering and meaningless.

FIRST FIVE NUMBERS (CHAPTER 3)	VALUE	READ
Trades per specification	Pullback 61 · Breakout 29	Neither meets any confidence threshold
Adherence rate	74%	Well below the 90% floor
Mean R / σ (all journaled)	+0.06R / 1.41	Indistinguishable from zero
Risk consistency (σ ÷ mean)	0.38	Sizing is being driven by conviction
R result vs currency result	+5.4R / -£1,180	Positive in R, negative in money

DRAWDOWN INSIGHT — THE FIFTH NUMBER

Positive in R and negative in currency is the signature of a sizing failure, and it is invisible to anyone tracking only one of the two. This trader has been losing money while their strategy has been marginally winning — the losses landed on the oversized positions. Nothing about their edge caused this.

Step 2 — Plateau diagnosis (Chapter 2)

Running the questions in order:

QUESTION	ANSWER
Could a stranger reproduce my trades from my written rules?	**No** — the pullback setup is written but includes "when momentum looks strong"; the breakout setup is not written at all

The diagnosis stops at plateau 1. The trader believed they were on plateau 5 — a decayed edge. They are two levels below it, and every conclusion they have drawn about their strategy over the past two months has been unsupported, because there was never a fixed strategy to draw conclusions about.

This is the single most common result of running the diagnostic honestly, and it is worth noting how far it sits from the trader's own account of the problem.

Step 3 — Fix specification, then re-measure

Six weeks of work: the pullback setup is decomposed until every condition is binary, "momentum looks strong" becoming three testable criteria. The breakout setup is written out properly, at which point the trader notices it is essentially the pullback setup with a looser entry — the same exposure, taken worse. It is retired rather than specified. Version 1.0 is dated and logged.

The trader then trades version 1.0 at minimum size for 60 trades, recording intent before outcome as Chapter 12 requires.

METRIC	BEFORE (JOURNALED 90)	AFTER 60 AT V1.0
Adherence	74%	93%
Risk consistency	0.38	0.11
Theoretical expectancy	not computable	+0.29R
Realised expectancy	+0.06R	+0.12R
Execution gap	not measurable	0.17R per trade

Two findings emerge immediately. The specification has a considerably better theoretical expectancy than anything the trader had previously measured — and a large execution gap is eating most of it. Neither fact was visible before specification, because there was nothing to compare behaviour against.

Step 4 — Decompose the gap (Chapter 16)

COMPONENT	COST PER TRADE	SHARE OF GAP	MECHANICAL CONTROL
Early exits	0.09R	53%	Take-profit placed at entry, no manual exits
Widened stops	0.04R	24%	Hard stop at entry; adjustable only in favour
Missed trades	0.02R	12%	Price alerts at setup levels
Slippage	0.02R	11%	Acceptable — structural, leave alone

Per the monthly framework, **one** control is installed first: the mechanical take-profit, targeting the largest component. The next month's data will show whether it worked, which it could not do if three controls were installed simultaneously.

Step 5 — Segment (Chapter 15)

With 60 clean trades the segments are small, so these are treated as hypotheses rather than findings — but two were predicted in advance and therefore carry more weight:

SEGMENT	TRADES	EXPECTANCY	STATUS
Trending regime	34	+0.34R	Predicted. Consistent with the edge rationale.
Ranging regime	26	-0.14R	Predicted. Candidate for a regime filter.
Post-loss trades	14	-0.31R	Sample far too small. Flag and re-test at 50.

The regime split has a mechanism behind it — a continuation setup should not pay in a market that isn't continuing — and it was hypothesised before the data was cut. It is adopted as version 1.1, and a fresh sample begins.

Step 6 — Size correctly (Chapter 8)

With realised expectancy around +0.20R and σ of 1.4 after the filter, full Kelly would suggest an implausible figure. The four constraints instead give: statistical (quarter Kelly, further reduced for a sample under 200) → 1.2%; drawdown (95th percentile within tolerance) → 0.8%; structural → none binding; behavioural (the size at which the trader stopped interfering, established during the minimum-size phase) → **0.5%**.

The smallest constraint governs. Risk is set at 0.5%, well below what the mathematics alone would permit — which is exactly the point of including a behavioural constraint at all.

Step 7 — Nine months on

METRIC	AT DIAGNOSIS	NINE MONTHS LATER
Trades of one specification	0	240 at v1.1
Adherence	74%	96%
Execution gap	0.17R	0.06R
Realised expectancy	+0.06R	+0.23R
95% confidence interval	crosses zero	+0.05R to +0.41R
Risk consistency	0.38	0.07
Plateau	1 — no specification	Cleared 1-4; monitoring for 5

The confidence interval now excludes zero. For the first time in three years, this trader can state — with evidence rather than hope — that they have an edge.

> DRAWDOWN INSIGHT — WHAT ACTUALLY CHANGED
>
> Nothing about the market. Nothing about the entry technique, which is the same pullback setup they were already trading. What changed was that the setup was specified, one redundant variant was retired, a regime filter was added on a predicted mechanism, exits were mechanised, and sizing was constrained by behaviour rather than ambition. The edge was always there. It was being spent on execution and hidden by measurement failure.

What this case is meant to demonstrate

- **The presenting complaint is rarely the problem.** "My edge stopped working" was, on inspection, "I have never had a fixed specification to measure."
- **The diagnostic order does the work.** Stopping at plateau 1 prevented two months of pointless strategy research from becoming six.
- **Most of the recovered performance came from execution**, not from strategy. The gap fell 0.11R; the specification never changed its entry logic.
- **Sample size discipline prevented two false conclusions** — the post-loss segment was flagged rather than acted on, and the early positive result was not sized up prematurely.
- **The timeline was nine months.** Not nine weeks. That is the realistic pace of this work, and expecting otherwise is itself a cause of failure.

> WARNING — THE SEDUCTIVE SHORTCUT
>
> The temptation on reading this is to skip to the interventions — mechanical exits, a regime filter, tighter sizing — and apply them directly. Don't. Those were the correct answers for this dataset, derived from measurement. Applied blind to a different trader they may be irrelevant or harmful. The transferable part is the sequence, not the conclusions.

KIT

THE TOOLKIT

Diagnostics, Tables & Templates

Print these. Everything here appears in the chapters — this is the working set, gathered for use.

TOOLKIT A

The Consistency Audit

Score honestly. Anything you cannot answer scores zero — an unmeasured item is an unmanaged one.

DIMENSION	0 POINTS	1 POINT	2 POINTS	SCORE
Specification	Rules in my head	Written, some judgement	A stranger could reproduce my trades	
Sample	<50 trades of one spec	50–200	Above my Chapter 6 requirement	
Adherence	Not measured	85–95%	>95%, measured every trade	
Risk consistency	Size varies with conviction	Mostly fixed	Formula-driven, σ/mean <0.2	
Expectancy	Never computed	Computed in aggregate	Per spec, net of costs, with CI	
Execution gap	Never measured	Known in total	Decomposed into five components	
Distribution	No idea what drawdown to expect	Rough sense	Simulated; know my 95th percentile	
Segmentation	Never done	Done once	Hypothesis-led, reviewed monthly	
Monitoring	Watch the balance	Track expectancy	Four decay metrics vs confidence band	
Rhythm	Ad hoc	Weekly review mostly happens	Daily/weekly/monthly/quarterly, with decision hierarchy	

16–20
OPERATING PROFESSIONALLY. FOCUS ON SCALING (CH 18–19)

9–15
THE APPARATUS IS PARTLY BUILT. COMPLETE THE LOWEST-SCORING DIMENSION FIRST

0–8
YOU CANNOT YET INTERPRET YOUR OWN RESULTS. START AT CHAPTER 3

TOOLKIT B

Six Plateaus Diagnostic

QUESTION — ANSWER IN ORDER	IF NO
Could a stranger identify my setups from my written rules alone?	**Plateau 1 — No specification** → Chapter 10
Do I have 100+ recorded trades of one unchanged specification?	**Plateau 2 — No sample** → Chapter 6
Is adherence above 90% across that sample?	**Plateau 3 — Execution gap** → Chapter 16
Is risk consistent, and do R and currency results agree?	**Plateau 4 — Sizing failure** → Chapter 8
Was expectancy once positive and is it now declining with adherence intact?	**Plateau 5 — Edge decay** (if YES) → Chapter 13
All the above in place and expectancy never positive?	**Plateau 6 — No edge** → Chapters 9 and 21

TOOLKIT C

Sample size reference

Trades required for a 95% confidence interval to exclude zero. Formula: $n \approx (1.96 \times \sigma \div E)^2$

WIN RATE	AVG WIN	EXPECTANCY	Σ	TRADES NEEDED
60%	1.0R	+0.20R	0.98	92
50%	1.5R	+0.25R	1.25	96
45%	1.5R	+0.125R	1.24	380
40%	2.0R	+0.20R	1.47	207
35%	2.0R	+0.05R	1.43	3,146
30%	3.0R	+0.20R	1.83	323

Assumes 1R losses and independent trades. Real requirements are higher — market outcomes cluster.

TOOLKIT D

Streak & drawdown reference

WIN RATE	TYPICAL LONGEST LOSING RUN (200 TRADES)	1 IN 20 REACHES	1 IN 100 REACHES
60%	5	8	10
50%	7	10	13
45%	8	12	15
40%	9	14	17
35%	11	16	20
30%	13	19	24

Maximum drawdown by risk level — System C (40% win, +2R), 200 trades, 20,000 simulations:

RISK / TRADE	MEDIAN MAX DRAWDOWN	95TH PERCENTILE MAX DRAWDOWN	MEDIAN FINAL EQUITY
0.5%	5.9%	11.0%	1.21×
1.0%	11.9%	21.0%	1.46×
2.0%	23.1%	38.3%	2.04×
3.0%	33.0%	52.1%	2.74×

Recovery required: −11% needs +12% · −21% needs +27% · −38% needs +62% · −52% needs +108%.

TOOLKIT E

Backtest log — column structure

COLUMN	NOTES
date · instrument · spec_version	Version matters — never mix versions in one sample
regime · volatility_state	Recorded at signal time, from rules, not impression
entry · stop · target · R_risked	As the specification dictates, not as hindsight suggests
exit_price · exit_reason	Target / stop / time / rule
result_R	Net of assumed costs
sample_segment	in-sample · validation · locked
ambiguous (Y/N)	Flag borderline signals. If >10% are ambiguous, the specification isn't finished.

TOOLKIT F

Segmentation template

State the hypothesis before computing. Compute the confidence interval for each segment separately.

SEGMENT	TRADES	EXPECTANCY	CI EXCLUDES ZERO?	MECHANISM?	ACTION
Setup × trending regime					
Setup × ranging regime					
Session 1 / Session 2					
Post-loss vs normal					
High vs low volatility					

TOOLKIT G

Edge health monitor

METRIC	HEALTHY	INVESTIGATE	ACT
Rolling expectancy (50)	Inside confidence band	Below lower bound <30 trades	Below lower bound 30+ trades → halve risk
Win rate, rolling	Within ±5pp of tested	5–10pp below	>10pp below sustained
Average win (R)	Within 15% of tested	15–25% below	>25% below → check volatility regime
Setup frequency	Within 20% of normal	20–40% below	>40% below → conditions have changed
Adherence	>95%	90–95%	<90% → drift, not decay. Fix execution first.
Execution gap	Stable or narrowing	Widening slightly	Widening materially → behavioural warning

Metrics reference

METRIC	FORMULA / DEFINITION
Expectancy (R)	(Win% × avg win) − (Loss% × avg loss)
Standard error	σ ÷ √n
95% confidence interval	E ± 1.96 × σ ÷ √n
Sample required	$(1.96 \times \sigma \div E)^2$
System quality	Expectancy ÷ σ — the number to optimise
Profit factor	Gross profit ÷ gross loss
Kelly fraction	f* = p − q ÷ b — then divide by 4 or more
Execution gap	mean(theoretical R − actual R)
Risk consistency	σ(risk per trade) ÷ mean(risk per trade) — target <0.2
Recovery required	1 ÷ (1 − drawdown) − 1
Annual return driver	Expectancy × frequency × risk per trade

Glossary

Adherence rate	Percentage of trades executed exactly as specified. The most predictive single metric available to a trader.
Bootstrap	Resampling your own recorded results with replacement to generate a distribution of possible outcomes.
Confidence interval	The range within which the true value plausibly sits, given your sample. If it crosses zero, you have not established an edge.
Curve fitting	Tuning parameters until they match historical noise. Produces excellent backtests and poor live results.
Decay	Genuine deterioration of an edge as conditions, participants or costs change.
Drift	Deterioration in results caused by the trader ceasing to execute the specification. Frequently misdiagnosed as decay.
Execution gap	Theoretical result minus realised result, per trade, in R.
Expectancy	Average R per trade. Determines whether repetition helps or harms.
In-sample / out-of-sample	Data used to develop a strategy, versus data reserved to test it. Only the latter is evidence.
Kelly criterion	The bet fraction maximising long-run geometric growth. A useful upper bound, not an instruction.
Max favourable excursion	How far a trade moved in your favour before exit. Reveals premature exits.
Monte Carlo	Simulating many possible sequences to examine the distribution of outcomes rather than one path.
Multiple comparisons	Testing many variants until one appears significant by chance. The main source of false discoveries.
Noise floor	The sample size below which results carry no information at all.
Plateau	One of six structural reasons competent traders stall (Chapter 2).
R / R-multiple	Result expressed as a multiple of the amount risked. The unit all analysis here uses.
Risk of ruin	Probability that variance exhausts capital before the edge pays. Rises non-linearly with risk per trade.
Sequence risk	The effect of trade order on the path — same trades, different order, materially different experience.
Specification	A rule set precise enough that a stranger reproduces your trades from it.
Walk-forward	Rolling optimisation and testing through history, aggregating only out-of-sample results.

WHAT HAPPENS NEXT

Keep going

This book gives you the analytical framework. What it cannot give you is the sample — that only comes from executing, recording and reviewing, for longer than feels reasonable.

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Foundation £49/mo	Building the fundamentals — structured education and the discipline framework.
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> **Consistency is not an outcome you achieve.**
> **It is a measurement capability you build.**
>
> DRAWDOWN